Cross-Outcome Falsification Tests with Few Clusters: Size Distortion and Randomization Calibrations

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Abstract

Falsification tests compare estimated effects across outcome categories differing in exposure to a suspected violation. Run as one joint test of m restrictions with cluster-robust standard errors, chi-square calibration is oversized: at $G = 51$ and $m = 13$ its null rejection rate is 0.389. We give three randomization calibrations and characterize each: label permutation, exact in finite samples under category exchangeability and needing no condition on G; cluster sign changes, exact under sign symmetry of the cluster contributions; and the restricted wild cluster bootstrap, simulated only. Permutation p-values floor at $2/C!$ for a two-sided statistic with exposure indices symmetric about their mean, so permutation needs $C \geq 5$ at the 5% level; at $C = 5$ its size-adjusted power is 0.383 against 0.983 for the directed cluster sign-change test. Under equal cluster design weights the Wald statistic is a scaled Hotelling T^2, which prices the distortion and bounds m; every design here violates that condition.

Keywords: randomization inference; cluster-robust inference; falsification tests; permutation tests; Hotelling's T^2; difference-in-differences.

JEL codes: C12, C15, C23.

*Charter School of Wilmington, Wilmington, DE. Email: `ethanxie@udel.edu`. Replication code, the simulation scripts and the output files underlying the tables are at `https://github.com/opxie1/Exact-Randomization-Inference-for-Cross-Outcome-Falsification-Tests-with-Few-Clusters`, on the `main` branch. The note to each table and figure names the output file its entries are read from. The simulations are written in Python and need version 3.11 or later with `numpy`, `scipy` and `pandas`. They draw pseudo-random numbers from the NumPy PCG64 generator, seeded once per replication by the CRC-32 checksum of a string naming the design cell and the replication index, so that any cell can be regenerated on its own. Producing every output file cited here takes 61.2 minutes in total (`results/theory/inference/summary.json`, field `runtime_min_total`).

1 Introduction

Falsification tests built from several outcomes are standard in applied work. The design is familiar: a researcher suspects that an estimated effect is contaminated by some mechanism, observes several outcome categories that differ in exposure to that mechanism, and tests whether the estimated effects line up with exposure. Iyer et al. (2012) separate reporting from incidence by contrasting documented crimes against women with crimes against men and gender-neutral crimes, and Miller and Segal (2019) run that contrast for violent victimization by sex of the victim. Weisburd et al. (2024) run a version of it across crime categories graded by reporting propensity. Baicker et al. (2013) report effects of Medicaid coverage on measured blood pressure and cholesterol next to self-reported diagnoses of the corresponding conditions. In epidemiology the device is formalized as the negative-control outcome (Lipsitch et al. 2010). Run as a single test across categories, the contrast is a joint restriction on interaction coefficients, and the p-value comes from a chi-square reference distribution with cluster-robust standard errors.

That calibration fails in the sample sizes where these tests are run. At $G = 51$ and $m = 13$ we measure a null rejection rate of 0.389 against a nominal 0.05, and at $G = 20$ the rate is 0.905. Section 5 prices the failure. When every cluster carries the same design weight, the cluster-robust Wald statistic is ω_2 times a Hotelling T^2 with dimension m and $G-1$ degrees of freedom (Hotelling 1931), where m is the number of restrictions, G is the number of clusters, and ω_2 is a fitting correction lying between 0.995 and 0.9996 across the designs of Section 9; the chi-square reference ignores the estimation of the $m \times m$ meat matrix from G cluster scores.

Equal design weights are what make that statement exact, and they are also what make the problem tractable. The condition fails in every design run below, including both applications, where G_{eff} of Definition 1 is 45.9, 36.0 and 32.8 against $G = 51$. Proposition 2 therefore enters as a diagnostic and as the source of a design rule, not as a calibration we recommend: its own reference distribution rejects at 0.082 to 0.175 in the twelve cells of Table 2. The calibrations we recommend are the randomization procedures, which do not use equal design weights and, in the case of Theorem 1, use no condition on G at all. The distortion is governed by m and G jointly, and the chi-square calibration is valid only when m grows slower than $G^{2/3}$. Along sequences with $m^{3/2}/G$ converging to a positive constant the size converges to a limit strictly above the nominal level, and when m/G converges to a positive fraction the size converges to one. This gives a design rule computable before any outcome is observed: at 36.4 effective clusters, size stays below 0.075 only for $m \leq 2$.

We give three alternative calibrations and characterize the size of each. The first permutes the exposure labels across outcome categories. Under an exchangeability condition on the joint law of the category data arrays, this test has size at most α in finite samples for every N, T, G and C, with no distributional, clustering or cluster-count assumption. The randomization

group is the set of $C!$ relabelings, and its order bounds the resolution: the smallest attainable p-value is $2/C!$ when the statistic is even in the labels and the exposure indices are symmetric about their mean, and $1/C!$ otherwise, so a two-sided test at level 0.05 with equally spaced indices requires $C \geq 5$. The second procedure randomizes sign changes over clusters, and it is exactly level α when the cluster contributions are sign-symmetric under the null of no cross-category gradient, asymptotically level α in G otherwise. The third is the restricted wild cluster bootstrap of Cameron et al. (2008), for which we report simulated size rather than a theorem. Both cluster-level procedures remain usable at $C \leq 4$.

Two results concern power rather than size. The Wald statistic is not optimal against the alternative these tests are designed to detect. When the shape of the response across horizons is known up to scale, a directed statistic is uniformly most powerful invariant, and recalibrating both tests to deliver size α puts the ratio of minimum detectable responses at power 0.80 at 0.393 at 36.4 effective clusters. Label randomization carries a price: relative to a normal benchmark the permutation test needs 1.35 times the signal at $C = 5$ and 1.15 at $C = 8$. Against a flat recording response of magnitude $\kappa = 0.45$ at $G = 51$ and $C = 5$, the label-permutation test with the two-sided Wald statistic attains size-adjusted power 0.383, while the cluster sign-change test applied to the directed statistic attains 0.983 and has a null rejection rate between 0.044 and 0.065 over $G \in \{20, 30, 51, 100\}$ and $C \in \{3, 5, 8\}$ at nominal level 0.05, with a Monte Carlo standard error of 0.005 from 2,000 replications. The label-permutation test fixes a reference distribution valid for every G under Assumption 4, and we use it as the finite-sample benchmark. The directed cluster sign-change test is the procedure we recommend running.

Section 4 states the design and the null. Section 5 derives the exact null law of the Wald statistic and the condition under which its chi-square calibration is valid. Section 6 gives the label-permutation test, its exactness condition, and its resolution floor. Section 7 gives the two cluster-level procedures and the sign symmetry condition under which the first is exactly level α. Section 8 derives local power and a closed-form minimum detectable response, and Theorem 3 gives the optimal statistic. Section 9 reports size and power over $G \in \{20, 30, 51, 100\}$ and $C \in \{3, 5, 8\}$. Section 10 applies the procedures to two panels. Appendix A contains the proofs.

2 Related work

Our exact null law is a statement about the cluster-robust Wald statistic with G clusters, which places it in the few-cluster inference literature. Cameron et al. (2008) propose the wild cluster bootstrap with the null imposed and document its size improvement over the cluster-robust t; MacKinnon and Webb (2017) characterize when it fails. Neither paper treats the joint m-restriction case in which m is comparable to G, which is where cross-outcome falsification

tests sit. Carter et al. (2017) show that cluster heterogeneity reduces the number of clusters that the asymptotics effectively deliver, and define the effective cluster count we use in Section 5. Canay et al. (2017) give asymptotic validity for a randomization test built on a group of fixed cardinality when the invariance holds only in the limit. The label-permutation group of Section 6 has cardinality $C!$, which does not vary with G, so their framework covers designs in which the exchangeability of Assumption 4 holds for the category statistics only as G grows. The finite-sample argument itself is the classical group-invariance construction, for which we follow Lehmann and Romano (2005).

Four further results enter directly. MacKinnon et al. (2023) set out current practice for cluster-robust inference, and Section 7 takes the restricted wild cluster bootstrap in the form given there. Ibragimov and Müller (2010) obtain a t-test on cluster-level estimates whose level is controlled with a small fixed number of heterogeneous clusters and no appeal to $G \to \infty$; that construction and the label-permutation test of Section 6 differ in what is treated as the repeated unit, clusters there and outcome categories here. Young (2019) recomputes p-values by randomization inference for a sample of published experimental papers in economics and reports that a substantial share of the rejections at conventional levels do not survive, which is the direction of the gap between the chi-square and the permutation p-value in Section 10. Djogbenou et al. (2019) prove a central limit theorem for the wild cluster bootstrap distribution conditional on the data, and that theorem supplies the asymptotic level of the cluster sign-change test of Section 7, which is the null-imposed Rademacher wild cluster bootstrap written in score form.

A small-sample cluster-robust literature already corrects both the variance estimator and the reference distribution for statistics of this kind. Bester et al. (2011) hold the number of clusters fixed and let cluster size grow, and obtain limiting t and F distributions in which G plays the role of the sample size; the $F(m, G-m)$ shape of (5.2) is theirs as a large-cluster approximation. Bell and McCaffrey (2002) remove the downward bias of the cluster-robust variance estimator under a working model and pair the correction with Satterthwaite degrees of freedom, and Pan and Wall (2002) do the same for generalized estimating equations. Imbens and Kolesár (2016) give degrees-of-freedom adjustments for the heteroskedasticity-robust case. For a hypothesis with $m > 1$ constraints, Tipton and Pustejovsky (2015) refer the statistic to an approximate Hotelling T^2 whose denominator degrees of freedom are matched from the data, and Pustejovsky and Tipton (2018) extend that test to models with arbitrary sets of fixed effects and report Type I error close to the nominal level across the designs they simulate.

Proposition 2 does not introduce that reference distribution. It states an algebraic identity between the CR1 Wald statistic and a Hotelling T^2, and, under Assumption 3, an exact $F(m, G-m)$ law for the joint null that the post-treatment coefficients do not vary with exposure, holding at every G and m with $G - m \geq 1$ rather than at denominator degrees of freedom matched from the data, and rather than in the large-cluster limit of Bester et al.

(2011). What is added is the finite-sample statement and the explicit constant ω_2, not the family. Corollary 2 is the second difference. It gives the rate $m = o(G^{2/3})$ at which the chi-square calibration stays valid, the limit the size converges to when $m^{3/2}/G$ tends to a positive constant, and the convergence of the size to one when m/G does. Calhoun (2011) derives the corresponding correction for the regression F test when the number of restrictions grows with the number of observations rather than with the number of clusters, and reports the same direction of failure for the uncorrected test.

The design that motivates the problem is the use of graded outcome categories to detect a treatment-induced change in how outcomes are recorded. Weisburd et al. (2024) run this contrast across crime categories. Hausman et al. (1998) formalize outcome misclassification that depends on treatment status in a cross-sectional discrete-response setting, and Denteh and Kédagni (2022) and Gutknecht and Liu (2024) study misclassified outcomes in difference-in-differences and changes-in-changes designs respectively. Freyaldenhoven et al. (2019) use an auxiliary series that loads on the same latent confound as the outcome and use it to purge event-study coefficients of that confound. The device is close to the normalization that Assumption 4 calls for here, and it is put to a different use: dividing each category outcome by an estimate of its loading on the common factor equalizes those loadings, which is what makes the category arrays exchangeable, and it removes nothing from the coefficient under test. We take the estimand and the identification argument from that literature as given. This paper is about the reference distribution.

Falsification tests are also run as pre-trend tests, where Roth (2022) shows that conditioning on passing one distorts the sampling distribution of the effect estimate, and Rambachan and Roth (2023) replace the test with bounds on the post-treatment violation. Our results concern the calibration of a cross-outcome test rather than the consequences of conditioning on its outcome, and the two problems compound: a test with size 0.39 conditions on an event that is mostly noise.

3 What the falsification test is for

That a policy can change how an outcome is recorded rather than the outcome itself is not a new observation. It is Campbell and Stanley (1963)'s *instrumentation* threat to internal validity, and their illustration is almost exactly the design studied below: "the experimental change of putting into office a new principal may produce a change in the recording of discipline infractions rather than in the infraction rate itself." Campbell (1969) makes the same point for policing, where a change in administration reforms the bookkeeping alongside the policy. Nothing in this section claims that observation.

What is claimed is a correction to what the classical tradition concluded about it. For the single-series design, Campbell and Stanley (1963) mark instrumentation as the one threat

the pre-period cannot dispel, while maturation, testing and regression are ruled out precisely because a pre-series is available. Then, for the multiple time-series design, which adds a control series and is the ancestor of the design used here, they write that "the difference in the selection of the groups operating in conjunction with maturation, instrumentation, or regression, can hardly account for an apparent effect." The reasoning is about instrument differences that arise from how the groups were selected. A recording response caused by the treatment is not such a difference. The control series cannot absorb it, because the control units do not receive the treatment that induces it. The distinction between an instrument difference across groups and a recording response induced by the treatment is not drawn there, and the applications cited below indicate that it is not always drawn now.

The consequence is sharper than the classical warning. When the contamination is a recording response, a pre-trend test and a sensitivity analysis calibrated from pre-trends are not merely low-powered against it. Neither carries any information about it at all. This section states that precisely, prices it, and marks the boundary of what it covers.

The gap is live. Reviewing difference-in-differences designs for pandemic policy, Goodman-Bacon and Marcus (2020) warn that estimates "will be biased if testing improves differently in the treatment and control groups," then recommend event-study estimates, which "can reveal many of the biases outlined above." Corollary 1 says that when the policy is what changed the recording, that remedy rejects with probability equal to its size. Richardson et al. (2025) report a formal pre-trend test as evidence that parallel trends holds, and separately concede that their estimated increase in reported intimate partner violence may reflect disclosure rather than incidence, without connecting the two. Baum et al. (2026) evaluate prenatal screening mandates, a treatment whose direct effect is on detection, and validate the design with a flat pre-period and a placebo-in-time test.

3.1 A recording response

Index units by i, periods by t, outcome categories by c, and adoption cohorts by g, with $\ell_{it} = t - G_i$ the event time and $D_{it} = \mathbf{1}\{t \geq G_i\}$. Write Y^*_{itc} for the latent outcome and Y_{itc} for what the recording system reports.

Assumption 1 (Separable recording response)**.** There exist μ_{ic}, depending arbitrarily on G_i and on the latent outcomes, deterministic ϕ_{tc}, and deterministic paths $m_c(\ell)$ with $m_c(\ell) = 0$ for $\ell < 0$, such that

$$Y_{itc} = Y^*_{itc} + \mu_{ic} + \phi_{tc} + m_c(\ell_{it})D_{it} + u_{itc}, \qquad E[u_{itc} \mid G_i] = 0. \tag{3.1}$$

Two features of Assumption 1 carry the results below and both are restrictions. The distortion enters additively and separably, so a recording map whose response depends on the level of the latent outcome is outside it. And $m_c(\ell) = 0$ for $\ell < 0$ is no anticipation in

the recording layer: the recording system changes when treatment arrives, not before. Unit-by-category terms μ_{ic} are unrestricted, so any time-invariant difference in recording practice across units is absorbed and only the treatment-induced change matters.

3.2 Pre-trend tests have power equal to size

Let $\Delta_c(g,t)$ denote the deviation from parallel trends in the latent layer. The following is a coupling argument: it exhibits, for any recording response, a structure with no recording response whatever that generates the same observed data.

Lemma 1 (Observational equivalence)**.** *Fix a structure satisfying Assumption 1 with latent parallel trends and recording paths m_c. The structure with no recording response and latent deviation*

$$\Delta_c(g,t) = m_c(t-g)\,\mathbf{1}\{t \geq g\} \tag{3.2}$$

generates the same law of the observed panel, cell by cell. Its pre-treatment deviations are exactly zero.

Corollary 1 (No pre-treatment diagnostic separates the two)**.** *Let φ be any test that is a measurable function of the observed panel and the cohort labels. Its rejection probability is the same under both structures. In particular a pre-trend test, a placebo-in-time test, and a leads-and-lags plot have rejection probability equal to size against a recording response.*

That pre-treatment parallelism fails to guarantee post-treatment parallelism is long established (Kahn-Lang and Lang 2020; Roth et al. 2023), and Ghanem et al. (2026) show pre-trend tests are uninformative in the presence of unrestricted structural breaks. Corollary 1 is the version of that statement for one named mechanism, and the mechanism supplies what the general statement cannot: the deviation in (3.2) is not merely unbounded by pre-period behaviour, it is identically zero there, by construction rather than by coincidence. A recording response is a structural break located exactly at the treatment date, which is the form Campbell and Stanley (1963) anticipated in words and which the results below put in a form that can be checked.

The construction resembles a familiar one and the resemblance is worth being precise about. Malani and Reif (2015) observe that anticipation effects and parallel-trends violations are observationally equivalent, and Fenaroli (2026) works out the consequences for sensitivity analysis. The direction is what differs. An anticipation effect moves the pre-period, which is why a pre-trend test can see it, and why the equivalence there is a problem about how to interpret a rejection. A recording response leaves the pre-period exactly as it was. There is nothing to interpret, because there is nothing to reject.

3.3 Sensitivity analysis calibrated from pre-trends

The standard response to Corollary 1 is to abandon testing and report a confidence set that is honest over a class of deviations (Rambachan and Roth 2023). Two classes are in common use. The smoothness class $\Delta^{\mathrm{SD}}(M)$ bounds second differences of the deviation path by M. The relative-magnitudes class bounds post-treatment first differences by a multiple of the largest pre-treatment first difference,

$$\Delta^{\mathrm{RM}}(\bar{M}) = \Big\{\delta : |\delta_{\ell+1} - \delta_\ell| \le \bar{M} \max_{s<0} |\delta_{s+1} - \delta_s| \ \ \forall \ell \ge 0\Big\}. \tag{3.3}$$

Substituting (3.2) into (3.3) gives the difficulty at once. The equivalent deviation is zero throughout the pre-period, so the quantity the class scales is zero, and $\Delta^{\mathrm{RM}}(\bar{M}) = \{0\}$ for every $\bar{M} \ge 0$. The family is not a nested sequence of relaxations indexed by $\bar{M}$; it is the point null repeated. There is no breakdown value, because no member of the family admits any deviation at all.

Proposition 1 (The relative-magnitudes family is empty against a recording response)**.** *Under Assumption 1 with $m_c \not\equiv 0$, the population identified set for the latent effect under $\Delta^{\mathrm{RM}}(\bar{M})$ is the singleton $\{\vartheta_c(\ell) + m_c(\ell)\}$ for every $\bar{M} \ge 0$, and it excludes the latent effect by exactly the recording wedge.*

Proposition 1 is a population statement and the finite-sample version is what a researcher meets. The honest set conditions on the estimated pre-period coefficients rather than on the true deviation, so the estimated bound is $\bar{M} \max_s |\widehat{\beta}_{s+1} - \widehat{\beta}_s|$, which is positive in every realized sample. Under a recording response that bound is pure sampling noise of order $n^{-1/2}$ while the wedge it is meant to cover is of order one. The interval therefore shrinks around the contaminated estimand at the sampling rate, and coverage of the latent effect falls to zero.

Arora and Chand (2026) note the mechanism in prose, cautioning that when pre-trends are uninformative Δ^{RM} produces "intervals whose width tends to zero even as the bias does not." Their remark is a paragraph in a paper about a different estimand, carries no formal statement, and treats flat pre-trends as a circumstance that may or may not obtain. Assumption 1 identifies a mechanism under which it obtains exactly, always, and undetectably.

A second reading deserves an answer. Mikhaeil and Harshaw (2025) describe the same collapse and treat it as correct behaviour, on the ground that an explicit conditional-extrapolation assumption licenses inferring small post-treatment violations from small pre-treatment ones. That defence is sound where the assumption holds. A recording response is precisely a violation of it that passes every pre-test by construction, so the assumption cannot be checked in the direction that matters here.

For the smoothness class the corresponding statement is weaker and older. That a researcher cannot calibrate M from the data and retain honest coverage is the non-adaptivity

result of Low (1997) and Cai and Low (2004), stated for econometric practice by Armstrong and Kolesár (2018) and Armstrong and Kolesár (2020), who write that "to maintain honesty over the whole function class, a researcher must choose M a priori, rather than attempting to use a data-driven method." Kolesár and Rothe (2018) put the asymmetry sharply: the data can bound the smoothness constant from below but never from above, and an honest interval needs an upper bound. Freeman (2026) restates the principle in a difference-in-differences framing. Nothing in this paper improves on any of that. What is new is the connection: $\Delta^{\mathrm{SD}}(M)$ is convex and centrosymmetric, so the non-adaptivity result applies to it directly, and Rambachan and Roth (2023) do not cite it. Their reference list contains Low (1995) for the computation of fixed-length intervals and Armstrong and Kolesár (2018, 2020) for their optimality, but neither Low (1997) nor Cai and Low (2004).

3.4 What coverage does

Table 1 reports coverage of the latent effect in a single-cohort design with a latent effect of 1.0 and a recording wedge of 0.6 at the target horizon. Confidence sets are computed with a reference implementation of Rambachan and Roth (2023) rather than by hand. For the smoothness class it computes the optimal fixed-length interval of their Section 4.1. For the relative-magnitudes class it solves the identified set by their Lemma 2.2 and extends it by a normal sampling term, a construction its own documentation calls conservative relative to the conditional hybrid intervals the paper recommends; a coverage failure computed this way therefore bounds the failure of the recommended interval. The row labelled $\widehat{M}$ calibrates the smoothness bound as the reference software's exported routine does, as the maximum over one-sided upper confidence bounds for the second differences of the observed pre-period coefficients. The row labelled institutional instead sets the bound from the size of the recording change, which is the alternative this section recommends.

The pattern in Table 1 is not a small-sample artefact that more data repairs. More data is what causes it. The calibration input falls at the sampling rate while the wedge does not move, so every relative-magnitudes interval eventually excludes the latent effect, and larger $\bar{M}$ only delays the crossing. The institutional bound, which never consults the pre-period, is unaffected.

3.5 Scope

Three limits bound the claims above. Assumption 1 restricts the recording map to enter additively with a path indexed by event time; a map that responds to the level of the latent outcome is not covered. Lemma 1 and Corollary 1 do not require latent parallel trends, but Proposition 1 is stated for a single adoption cohort, and the extension to staggered designs is not proved here. And the argument is about what pre-treatment data can reveal; it says nothing

Table 1: Coverage of the latent effect by honest confidence sets under a recording response. Single cohort, latent effect 1.0, recording wedge 0.6 at the target horizon, nominal 95%. Sets computed with a reference implementation of Rambachan and Roth (2023). $\widehat{M}$ calibrates the smoothness bound from the pre-period as the reference software's exported routine does; institutional sets it from the size of the recording change. The last column is coverage of the recorded effect, which is what the interval is built to cover.

			$\Delta^{\mathrm{RM}}(\bar{M})$			$\Delta^{\mathrm{SD}}(M)$		
N	reps	$\widehat{M}$	$\bar{M}=0.5$	$\bar{M}=1$	$\bar{M}=2$	$M=\widehat{M}$	institutional	recorded
400	400	0.7165	0.680	0.955	0.998	1.000	1.000	1.000
1,600	400	0.3615	0.048	0.427	0.873	1.000	1.000	1.000
6,400	400	0.1814	0.000	0.007	0.287	1.000	1.000	1.000
25,600	200	0.0890	0.000	0.000	0.000	1.000	1.000	1.000
102,400	100	0.0450	0.000	0.000	0.000	0.210	1.000	1.000
409,600	20	0.0226	0.000	0.000	0.000	0.000	1.000	1.000

against sensitivity analysis whose bound comes from outside the event-study coefficients, which is the practice the last row of Table 1 supports and which Armstrong and Kolesár (2020) already recommend.

What survives is a design problem. If the contamination is a recording response, the pre-period is silent, and information has to come from somewhere else. The remainder of this paper takes it from the cross-section of outcome categories, which differ in how much discretion the recording system has over them, and addresses the inference problem that arises when the number of clusters is small.

4 Setup and the null

Units $i = 1, \dots, N$ are grouped into clusters $g = 1, \dots, G$; periods are $t = 1, \dots, T$; outcome categories are $c = 1, \dots, C$ with known exposure indices $d_1, \dots, d_C$. Treatment is absorbing with cohort $G_i \in \{1, \dots, T\} \cup \{\infty\}$ and $D_{it} = \mathbf{1}\{t \geq G_i\}$.

Assumption 2 (category-balanced design)**.** There is a common cell set $\mathcal{C} \subseteq \{1, \dots, N\} \times \{1, \dots, T\}$ with $|\mathcal{C}| = n_0$ on which every category is observed, and the treatment path D_{it}, the cohort G_i and the cluster map $i \mapsto g(i)$ do not depend on c.

Assumption 2 rules out designs in which categories are measured on different subsamples or at different frequencies. It is verifiable from the data layout. It is a hypothesis of Lemma 2 rather than of Theorem 1: when it fails, $\hat{\gamma}$ in (4.1) is no longer the fixed linear combination of per-category event studies that Lemma 2 gives and each category has to be fitted on its own cells, while the randomization of Section 6 acts in the category direction and needs only

Assumption 4.[1] Section 10 reports one application in which Assumption 2 fails through a cell pattern that depends on the category label, which breaks Assumption 4 as well.

Let L collect the included event-time horizons with $|L| = q$, omitting $\ell = -1$ and binning endpoints, let $X \in \mathbb{R}^{n_0 \times q}$ hold the corresponding indicators, and let $P \subseteq \{1, \ldots, q\}$ with $|P| = m$ be the post-treatment block. With y_{itc} the outcome and $\delta_c = d_c - \bar{d}$ the centered exposure index, the pooled gradient regression is

$$y_{itc} = a_{ic} + p_{tc} + \sum_{\ell \in L} \beta_\ell X_{(it),\ell} + \sum_{\ell \in L} \gamma_\ell \delta_c X_{(it),\ell} + u_{itc}, \tag{4.1}$$

estimated by OLS with the CR1 cluster-robust covariance clustered on g. The hypothesis under test throughout is the joint null

$$H_0 : \gamma_P = 0, \tag{4.2}$$

that the post-treatment coefficients do not vary with exposure. This is a weak null on a parameter vector, not a sharp null of no effect for every unit, and the distinction matters for the randomization arguments in Sections 6 and 7: we state separately for each procedure which invariance delivers validity and under what condition.

The cross-category structure enters (4.1) as a single interaction contrast. The procedure tests one joint hypothesis, not C separate hypotheses, so no multiplicity adjustment arises and the error rate controlled is the per-comparison rate for (4.2).

Write Π for the two-way within projection on $\mathcal{C}$, $\tilde{X} = \Pi X$, $S = \tilde{X}'\tilde{X}$, and $b_c = S^{-1}\tilde{X}'\Pi y_c$ for the within-category event-study coefficient vector.

Lemma 2 (separability)**.** *Under Assumption 2 and* $\operatorname{rank}(\tilde{X}) = q$*, the OLS estimators in* (4.1) *satisfy* $\hat{\beta} = C^{-1} \sum_c b_c$ *and* $\hat{\gamma} = \sum_c \delta_c b_c / \sum_c \delta_c^2$.

The gradient estimator is a fixed linear combination of the per-category event studies, which is what makes the category-level randomization of Section 6 available.

The b_c are within-category event studies estimated by OLS, which under staggered adoption with heterogeneous effects need not recover a convex combination of cohort-specific effects (de Chaisemartin and D'Haultfœuille 2020; Sun and Abraham 2021). Replacing each b_c by a heterogeneity-robust estimator (Callaway and Sant'Anna 2021; Borusyak et al. 2024), or by another of the alternatives surveyed in Roth et al. (2023), leaves the randomization arguments

[1] Fitting each category on its own cells, the label-permutation test of (4.2) rejects at 0.047 when the missing cells are tied to the category label and at 0.049 when they are independent of it, against a nominal 0.05 over 1,200 replications with a Monte Carlo standard error of 0.006 (`verification_extra6_3.json` in `results/theory/inference/`). With the observation rate falling from 1.00 to 0.40 across categories ordered by d_c, the rate is 0.040, against 0.053 for a control that assigns the same five rates to a category drawn at random in each replication, over 1,500 replications with a Monte Carlo standard error of 0.006 (`verification_extra7_8.json`, same directory). The second departure is downward, so the test stays at or below its nominal level and the cost falls on power.

of Sections 6 and 7 intact, because those require only that the category-level statistic be order-invariant in the sense of Theorem 1. The exact null law of Section 5 is specific to the CR1 Wald statistic from (4.1) and would have to be re-derived for another estimator.

Failing to reject (4.2) does not establish that the identifying assumption holds. Section 8 gives the minimum detectable response for each design, and the applications in Section 10 report it alongside every non-rejection.

5 The exact null law of the cluster-robust Wald statistic

The size distortion has a closed form. We isolate it under an idealization that is used only for this diagnosis and is not maintained elsewhere in the paper.

Let $\tilde{X}_g$ collect the rows of $\tilde{X}$ belonging to cluster g, let u_{gc} be the corresponding errors in (4.1), let $e_{gc} = \tilde{y}_{gc} - \tilde{X}_g b_c$ be the within-category residuals, and let $n = Cn_0$ be the number of observations in (4.1). The cluster score of the gradient block and its fitted counterpart are

$$a_g = \frac{\tilde{X}_g' \sum_c \delta_c u_{gc}}{\sum_c \delta_c^2}, \qquad a_g^f = \frac{\tilde{X}_g' \sum_c \delta_c e_{gc}}{\sum_c \delta_c^2}, \qquad \eta_g = S^{-1} a_g, \qquad \eta_g^f = S^{-1} a_g^f, \tag{5.1}$$

all in $\mathbb{R}^q$. Lemma 2 gives $\hat{\gamma} - \gamma = \sum_g \eta_g$, and the CR1 covariance of $\hat{\gamma}$ is $\omega \sum_g \eta_g^f \eta_g^{f\prime}$ with $\omega = \{G/(G-1)\}\{(n-1)/(n-2q)\}$. A subscript P denotes the subvector or submatrix indexed by the post block, and $\bar{\eta}_P = G^{-1} \sum_g \eta_{gP}$.

Assumption 3 (ideal cluster experiment)**.** (i) Under H_0 in (4.2), the cluster score contributions $\eta_1, \ldots, \eta_G$ are i.i.d. $N(0, \Sigma)$ with $\Sigma_{PP} \succ 0$. (ii) Every cluster carries the same design weight, $S_g = S/G$ for all g, where $S_g = \tilde{X}_g' \tilde{X}_g$.

Part (i) holds when the errors in (4.1) are Gaussian, independent across clusters, and identically distributed across clusters given the design. Part (ii) fails in a staggered design with a large never-treated share; Definition 1 handles that case.

Proposition 2 (exact null law)**.** *Let Assumptions 2 and 3 hold, let* $\operatorname{rank}(\tilde{X}) = q$*, let* H_0 *in* (4.2) *hold, and let* $G - m \geq 1$*. Put* $\omega_2 = (n-2q)/(n-1) = G/\{(G-1)\omega\}$*, let* $\hat{S}_P$ *be the sample covariance of the* η_{gP} *and* $T^2 = G\bar{\eta}_P' \hat{S}_P^{-1} \bar{\eta}_P$*. Then the CR1 Wald statistic* W *for* (4.2) *satisfies* $W = \omega_2 T^2$ *identically, and*

$$\frac{(G-m)\, W}{\omega_2 (G-1) m} \sim F(m, G-m) \tag{5.2}$$

exactly for every G *and* m *with* $G - m \geq 1$*. If instead* $G - 1 < m$*, the CR1 meat* $\sum_g \eta_{gP}^f \eta_{gP}^{f\prime}$ *has rank at most* $G-1$*, the Moore–Penrose version of* W *has no Hotelling representation, and* (5.2) *is void.*

The chi-square calibration treats the estimated meat matrix as known. Under Assumption 3 the cost of that substitution is exactly the F denominator in (5.2), which has $G-m$ degrees of freedom. The statement is a degrees-of-freedom result: nothing about the errors beyond Assumption 3(i) enters, and ω_2 equals 0.9989 at the headline design of Section 9 ($G = 51$, $C = 5$). The factor $(n-2q)/(n-1)$ counts the $2q$ interacted event-time regressors in (4.1) and not the $C(N+T-1)$ absorbed unit-by-category and period-by-category effects, so a user calibrating through (5.2) must supply the ω_2 that matches the degrees-of-freedom convention of their own CR1 implementation.

Inverting (5.2) for a critical value gives a reference distribution that is exact for the null (4.2) under Assumption 3, whose part (ii) requires every cluster to carry the same design weight. Where that part fails the calibration is neither exact nor conservative. In the staggered design of Section 9 with $C = 5$, referring W to (5.2) at the true G gives null rejection rates of 0.174, 0.160, 0.124 and 0.092 at $G = 20$, 30, 51 and 100 against the nominal 0.05, over 2,000 replications per cell (`anatomy_table.csv`, which is also the source for ω_2 above). The version evaluated at the G_{eff} of Definition 1 is an approximation whose accuracy Section 9 measures. The F family in (5.2) is the one that Tipton and Pustejovsky (2015) and Pustejovsky and Tipton (2018) reach by matching denominator degrees of freedom to the data; under Assumption 3 no matching is needed, because (5.2) holds exactly with $G-m$.

Corollary 2 (when the chi-square calibration is valid)**.** *Let* $\text{size}(G,m)$ *denote the null rejection probability of the nominal level-*α *chi-square test under the conditions of Proposition 2. Assume* $G(1-\omega_2) = O(1)$ *in part (i) and* $\sqrt{m}\,|\log\omega_2| \to 0$ *in part (ii), where* $1-\omega_2 = (2q-1)/(n-1)$. *(i) For fixed* m, $\text{size}(G,m) = \alpha + c_m f_m(c_m)\big[(m+c_m)/\{2(G-1)\} - (1-\omega_2)\big] + O(G^{-2})$ *where* $c_m = \chi^2_{m,1-\alpha}$ *and* f_m *is the* χ^2_m *density, so* $\text{size}(G,m) \to \alpha$ *as* $G \to \infty$. *The correction is positive, and the chi-square test over-rejects at all large* G, *when* $2(G-1)(1-\omega_2) < m + c_m$. *(ii) Let* $m \to \infty$ *with* $m/G \to 0$ *and* $\chi_{m,G} = \sqrt{m/2}\,\log\{(G-1)/(G-m)\} \to \chi \in [0,\infty]$. *Then* $\text{size}(G,m) \to \Phi(\chi - z_{1-\alpha})$, *and* $\chi_{m,G} = m^{3/2}/(\sqrt{2}\,G) + o(1)$ *whenever* $m^{5/2}/G^2 \to 0$. *If instead* $m/G \to r \in (0,1)$, *then* $\text{size}(G,m) \to 1$.

Part (ii) is the operative statement. The chi-square calibration is asymptotically valid if and only if $m^{3/2}/G \to 0$, that is $m = o(G^{2/3})$.

Figure 1 plots the exact size implied by (5.2) against m at five cluster counts.

The limit in Corollary 2(ii) is not accurate at the m this paper works with: along the sequence with $\chi = 0.5$, the exact size from (5.2) exceeds $\Phi(\chi - z_{1-\alpha})$ by 5.95 percentage points at $m = 13$, 4.21 at $m = 25$ and 2.91 at $m = 50$, an error falling like $m^{-1/2}$ (`anatomy_asymptotics.csv`). Every admissible-m figure reported below is computed from (5.2) and not from the limit.

Definition 1 (effective clusters)**.** Let J_P be the $q \times m$ selection matrix whose columns are the standard basis vectors indexed by P, so that $[S^{-1}]_{PP} = J_P' S^{-1} J_P$. For a contrast $w \in \mathbb{R}^m$ put

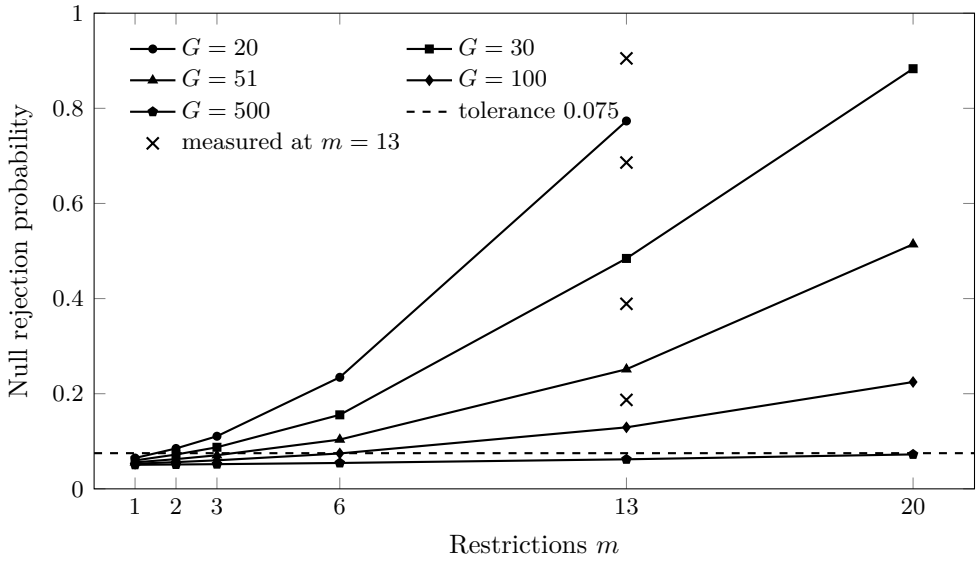


Figure 1: Exact size of the nominal 5% chi-square test against the number of restrictions

Note: each curve gives the rejection probability of the nominal level-0.05 chi-square test of the joint null $H_0 : \gamma_P = 0$ in (4.2), computed from (5.2) under Assumptions 2 and 3 at $\omega_2 = 1$, against the number of restrictions m, at five cluster counts G. The dashed line marks a tolerance of 0.075. The crosses are the measured rejection rates at $m = 13$ and $C = 5$ from Table 2, at $G = 20$, 30, 51 and 100 reading down, over 2,000 replications each with a Monte Carlo standard error of at most 0.011. They lie above the corresponding curves because the simulated design gives $G_{\text{eff}} = 0.727\,G$ rather than the equal cluster weights of Assumption 3(ii). Sources: `results/theory/inference/anatomy_expansion.csv`, `results/theory/inference/size_table.csv` and `results/theory/inference/anatomy_effective_clusters.csv`.

$a = S^{-1} J_P w$ and $\varphi_g = a' S_g a \geq 0$. Then $G_{\text{eff}}(w) = (\sum_g \varphi_g)^2 / \sum_g \varphi_g^2$.

Under errors that are homoskedastic and independent across clusters, $\varphi_g / \sum_h \varphi_h$ is the share of $\text{Var}(w'\hat{\gamma}_P)$ carried by cluster g. Write $\bar{\varphi} = G^{-1} \sum_g \varphi_g$ and $\Gamma = G^{-1} \sum_g \{(\varphi_g - \bar{\varphi})/\bar{\varphi}\}^2$. Then $G/(1+\Gamma) = G\bar{\varphi}^2 / (G^{-1} \sum_g \varphi_g^2) = (\sum_g \varphi_g)^2 / \sum_g \varphi_g^2$, so Definition 1 is the count $G/(1+\Gamma)$ of Carter et al. (2017) evaluated at the design weights φ_g. That paper defines Γ for the t-statistic on a scalar contrast through cluster-level contributions to the variance of the estimator; Definition 1 fixes those contributions at their homoskedastic design values, which is what makes it computable before any outcome is observed. It satisfies $1 \leq G_{\text{eff}}(w) \leq G$, with equality on the right when the φ_g are equal, which is what Assumption 3(ii) imposes. Substituting G_{eff} for G in (5.2) is an approximation; Proposition 2 does not license it. Section 9 reports its accuracy.

Every G_{eff} quoted below is the equal-weight contrast $w = \mathbf{1}_m$, and the design rule should be read at $\min_w G_{\text{eff}}(w)$ over the eigen-directions of $[S^{-1}]_{PP}$. The two differ little in the designs run here. At the $G = 51$, $C = 5$, $m = 13$ design of Section 9, $G_{\text{eff}}(w)$ ranges from 35.92 to 36.42 over those directions, the size that (5.2) predicts for the nominal 0.05 chi-square test at $m = 13$ ranges from 0.378 to 0.385, and the largest m holding that predicted size below 0.075

is 2 at either end (`geff_eigen_sweep.csv`). Both G_{eff} and m are computable from the design alone, which turns Corollary 2 into an ex-ante restriction on how many horizons a researcher may test.

6 The label-permutation test

Write $\mathcal{Y}_c \in \mathbb{R}^{n_0}$ for the category-c data array, $\mathcal{D} = (\mathcal{Y}_1, \ldots, \mathcal{Y}_C)$, and $\mathcal{Z}$ for the design, which is common to all categories under Assumption 2. Let $\mathcal{T}(\mathcal{D}, \delta)$ be a statistic with large values evidence against H_0 in (4.2).

The statistic is computed on loading-normalized arrays. Let $\hat{\Sigma}_{ck}$ be the sample covariance of the two-way demeaned categories c and k over the untreated cells $\{(i,t) \in \mathcal{C} : D_{it} = 0\}$, and put $L_{ck} = \log|\hat{\Sigma}_{ck}|$. The log-triad loading estimate is $\hat{\rho}_c \propto \exp(\hat{a}_c)$, scaled to geometric mean one, with

$$\hat{a}_c = \frac{1}{C-1}\sum_{k \neq c} L_{ck} \;-\; \frac{1}{2(C-1)(C-2)} \sum_{k \neq l,\; k,l \neq c} L_{kl}, \qquad C \geq 3. \tag{6.1}$$

Under a one-factor structure with $\hat{\Sigma}_{ck} \to_p \theta_c \theta_k s^2$ for $c \neq k$, the limit of L_{ck} is $\log|\theta_c| + \log|\theta_k| + \log s^2$, and the two averages in (6.1) isolate $\log|\theta_c|$ up to a constant common to all c, so $\hat{\rho}_c \to_p |\theta_c| / (\prod_j |\theta_j|)^{1/C}$. There is no reference category and no ordering convention in (6.1), which is what Lemma 3 records.

Definition 2 (label-permutation test)**.** Normalize each array, $\mathcal{Y}_c^\dagger = \mathcal{Y}_c / \hat{\rho}_c$, and drop every category with $|\hat{\rho}_c| \leq \rho_{\min}$, where $\rho_{\min} = 0.1$ throughout; write C_0 for the count of Section 4, C for the number of categories that survive the screen, and $\mathcal{D}^\dagger$ for the surviving normalized arrays. For $\pi \in S_C$ let $\delta_\pi = (\delta_{\pi(1)}, \ldots, \delta_{\pi(C)})$ and $T(\pi) = \mathcal{T}(\mathcal{D}^\dagger, \delta_\pi)$. With reference set $\mathcal{R} = S_C$ and $M = C!$, the test rejects at level α when $\hat{p} = M^{-1}\#\{\pi \in \mathcal{R} : T(\pi) \geq T(\mathrm{id})\} \leq \alpha$. When $C! > B$ the reference set is instead $\mathcal{R} = \{\mathrm{id}, \pi_2, \ldots, \pi_B\}$, with $\pi_2, \ldots, \pi_B$ drawn independently and uniformly from S_C, and $M = B$.

The randomization is over the assignments of the exposure labels to the normalized category data arrays, holding the arrays and the design fixed. Nothing is permuted at the unit, cluster or period level. Every run reported below sets $B = 999$, so the reference set is all of S_C at $C \leq 6$ and a Monte Carlo sample at $C = 8$.

Lemma 3 (equivariance of the normalization)**.** *Under Assumption 2, relabelling the arrays relabels the loading estimates identically: writing $\mathcal{D}_\sigma = (\mathcal{Y}_{\sigma(1)}, \ldots, \mathcal{Y}_{\sigma(C)})$, one has $\hat{\rho}_c(\mathcal{D}_\sigma) = \hat{\rho}_{\sigma(c)}(\mathcal{D})$ for every $\sigma \in S_C$. The retained set $\{c : |\hat{\rho}_c| > \rho_{\min}\}$ is relabelled with them, so $\mathcal{D} \mapsto \mathcal{D}^\dagger$ is S_C-equivariant.*

A statistic that is order-invariant in the normalized arrays is therefore order-invariant in the raw arrays, and Theorem 1 below applies to $\mathcal{D}^\dagger$ unchanged.

Two conditions scope (6.1), and both restrict the procedure rather than the proof. First, L_{ck} discards the sign of $\hat{\Sigma}_{ck}$, so (6.1) estimates $|\theta_c|$: dividing a negatively loaded category by a positive $\hat{\rho}_c$ leaves its factor component with the wrong sign, and the exchangeability of Assumption 4 below then fails at every sample size. The maintained condition is that the loadings share a sign, which is checkable from the observed signs of the $\hat{\Sigma}_{ck}$. Second, the estimator is unstable at a loading near zero. At $\theta = (0.5, 0.8, 0.02, 1.4, 2.0)$ the largest error in $\hat{\rho}$ is 1.18 at 2,000 untreated cells and still 0.34 at 4×10^4, against 0.009 at 4×10^4 when the smallest loading is 0.5; the guard drops such a category rather than normalizing by it. The guard did not fire in any run reported here: over 400 replications in each of the twelve (G, C) cells of Section 9 the smallest $|\hat{\rho}_c|$ is 0.539. Sources: `verification.json` for the loading errors and `loading_guard_check.csv` for the guard.

Assumption 4 (category exchangeability)**.** Conditional on $\mathcal{Z}$,

$$(\mathcal{Y}_{\pi(1)}, \ldots, \mathcal{Y}_{\pi(C)}) \stackrel{d}{=} (\mathcal{Y}_1, \ldots, \mathcal{Y}_C) \quad \text{for every } \pi \in S_C,$$

and $d_1, \ldots, d_C$ are fixed constants.

Assumption 4 rules out categories with different error variances and any dependence of the labels on the data. It restricts the category direction only: heteroskedasticity across units and periods, serial correlation, and within-cluster dependence of any form are unrestricted, and G is unrestricted. Normalizing the outcome does not restore it under fixed heterogeneous loadings. Dividing category c by its true loading θ_c equalizes the factor component and leaves $\mathrm{Var}(\varepsilon_c/\theta_c) = \sigma^2/\theta_c^2$ category-specific, and at $C = 5$ with $\theta = (0.4, 0.7, 1.0, 1.3, 1.6)$ the label-permutation test of Definition 2 run on the true-loading normalization has size 0.030 against the nominal 0.05, with Monte Carlo standard error 0.006 (`verification.json`, entry `V24`). Exchangeability does hold when the loadings are exchangeable draws independent of d, which is what the simulation of Section 9 imposes, and by Lemma 3 estimating them through (6.1) preserves it. Assumption 4 is not implied by H_0 in (4.2) and it is not directly testable, though its leading failure mode has an observable signature: Section 9 prices the size distortion as a function of the ratio of the largest to smallest category error variance.

Theorem 1 (exact finite-sample validity)**.** *Suppose Assumption 4 holds and $\mathcal{T}$ is order-invariant, in the sense that it is a function of the multiset $\{(\delta_c, \mathcal{Y}_c^{\dagger})\}_{c=1}^{C}$ and not of the order in which the pairs are listed. Then for every N, T, G and C, the test of Definition 2 for the null* (4.2) *under the randomization over its reference set $\mathcal{R}$ satisfies* $\Pr(\textit{reject}) \leq \alpha$. *If the distinct values among $\{T(\pi) : \pi \in \mathcal{R}\}$ are almost surely attained with a common multiplicity $\varkappa$, then* $\Pr(\textit{reject}) = \varkappa \lfloor \alpha M/\varkappa \rfloor / M$. *Here $\varkappa = 2$ for the two-sided Wald statistic with $\{\delta_c\}$ symmetric about zero, by Proposition 3(ii), and $\varkappa = 1$ for the one-sided directed statistic. At $C = 5$ and $\alpha = 0.05$ the two give* 0.05 *alike.*

Exactness here is finite-sample exactness for the null (4.2) under the randomization distribution of Definition 2, and it requires Assumption 4. No condition on G, on the error distribution, or on the within-cluster dependence structure is used. Order-invariance holds for the OLS estimates and CR1 covariance in (4.1), and by Lemma 3 it survives the normalization of Definition 2. It fails when the normalization is not S_C-equivariant, which a reference-category rule keyed to $\arg\min_c d_c$ is not. A position-keyed rule that acts as an order-invariant rescaling times a scalar common to all categories remains valid, because $\hat{\gamma}$ and its CR1 covariance in (4.1) are homogeneous of degree one and two in the outcome and the Wald statistic is therefore unchanged by a common rescaling: a normalization whose reference column is chosen by the position of the largest permuted label has size 0.052 against the nominal 0.05, with Monte Carlo standard error 0.005 (`verification.json`).

A randomized version has size exactly α. It rejects with probability one when $\hat{p} \leq \alpha_-$ and with probability $(\alpha - \alpha_-)/(\alpha_+ - \alpha_-)$ when $\hat{p} = \alpha_+$, where α_- is the largest attainable value of $\hat{p}$ not exceeding α and α_+ the smallest value exceeding it. No procedure reported below uses it.

Proposition 3 (attainable p-values)**.** *(i) On the exhaustive reference set, $\hat{p} \geq |\mathrm{Stab}(\delta)|/C!$, where $\mathrm{Stab}(\delta) = \{\pi \in S_C : \delta_\pi = \delta\}$ has order the product of the factorials of the sizes of the tied groups of $d_1, \ldots, d_C$, so the bound is $1/C!$ when d has no ties. On the Monte Carlo reference set of Definition 2, $\hat{p} \geq 1/B$. (ii) If $\mathcal{T}$ is even in the labels, so that $\mathcal{T}(\mathcal{D}^\dagger, -\delta) = \mathcal{T}(\mathcal{D}^\dagger, \delta)$, and the multiset $\{\delta_c\}$ is symmetric about zero and not identically zero, then $\hat{p} \geq 2|\mathrm{Stab}(\delta)|/C!$ on the exhaustive reference set. Equally spaced d is the leading sufficient case for the symmetry. (iii) On the exhaustive reference set with d untied, a level-α label-permutation test therefore requires $C! \geq 2/\alpha$ for a statistic even in the labels with $\{\delta_c\}$ symmetric about zero, that is $C \geq 5$ at $\alpha = 0.05$, and $C! \geq 1/\alpha$ otherwise, that is $C \geq 4$.*

The Wald statistic is even in the labels because $\hat{\gamma}(-\delta) = -\hat{\gamma}(\delta)$ and its covariance estimator is quadratic in the cluster scores. Proposition 3 is a property of the randomization group, not of any implementation. With $C = 3$ the two-sided Wald has three attainable p-values, the smallest being $1/3$, and the test can never reject at conventional levels. Section 9 reports the resulting rejection rate of exactly 0.000.

7 Cluster-level randomization and the bootstrap

Two procedures remain available when $C \leq 4$. Both randomize over clusters rather than category labels, so Proposition 3 does not bind.

The first randomizes sign changes over $\mathcal{E} = \{-1, +1\}^G$. Let γ^r be the restricted estimate, with $\gamma^r_P = 0$ and $\gamma^r_{\mathrm{pre}} = (S_{\mathrm{pre,pre}})^{-1}(S\hat{\gamma})_{\mathrm{pre}}$, let $z^r_g = \sum_c \delta_c \tilde{X}'_g\{\tilde{y}_{gc} - \tilde{X}_g(\hat{\beta} + \delta_c\gamma^r)\}$ be the

cluster-g score of the gradient block at that restricted fit, and put

$$\zeta_g = \frac{(S^{-1} z_g^r)_P}{\sum_c \delta_c^2} \in \mathbb{R}^m, \qquad \sum_g \zeta_g = \hat{\gamma}_P. \tag{7.1}$$

The test randomizes $\mathcal{Q}(\epsilon) = (\sum_g \epsilon_g \zeta_g)' \hat{\Omega}^{-1} (\sum_g \epsilon_g \zeta_g)$ over $\epsilon \in \mathcal{E}$, with $\hat{\Omega}$ proportional to $\sum_g \zeta_g \zeta_g'$, and rejects when $\mathcal{Q}(\mathbf{1}) = \hat{\gamma}_P' \hat{\Omega}^{-1} \hat{\gamma}_P$ falls in the upper α tail of $\{\mathcal{Q}(\epsilon)\}$. The CR1 scale factor in $\hat{\Omega}$ is common to every ϵ and cancels from the comparison, and $\hat{\Omega}$ itself does not depend on ϵ because $\epsilon_g^2 = 1$. Write ζ_g^0 for the same functional of the true errors under (4.2).

Theorem 2 (cluster sign-change test)**.** *If, under H_0 in* (4.2)*, the cluster contributions are jointly sign-symmetric, meaning $(\epsilon_1 \zeta_1, \ldots, \epsilon_G \zeta_G) \overset{d}{=} (\zeta_1, \ldots, \zeta_G)$ for every $\epsilon \in \mathcal{E}$, then the sign-change test is exactly level α in finite samples, and so is its Monte Carlo version using $B - 1$ random sign vectors together with the identity. Sign symmetry holds for the infeasible statistic built from the true errors under* (4.2) *whenever the cluster error contributions are independent across g and symmetric about zero. Under Assumption 3(ii) the feasible and infeasible contributions coincide exactly on P, so the feasible test is then exactly level α in finite samples. The feasible statistic differs from the infeasible one by $O_p(G^{-1/2})$ and is asymptotically level α as $G \to \infty$ under conditions (W1) to (W3) of Appendix A.*

Under Assumption 3(ii) the gap between the two statistics is zero, because $\zeta_g - \zeta_g^0 = -(S^{-1} S_g \gamma^r)_P = -\gamma_P^r / G = 0$ by construction of γ^r, and the measured gap is exactly zero at $G = 12$, 24, 48, 96 and 192 (`verification.json`). Outside that condition the finite-sample statement applies to the infeasible statistic alone, and the feasible statistic inherits validity only asymptotically in G. The argument for that limit in Appendix A is a conditional bootstrap central limit theorem rather than an approximate-symmetry argument, because the sign-change group has cardinality 2^G. Section 9 measures the finite-sample gap.

The statistic $\mathcal{Q}$ is even in ϵ, so its 2^G values pair and the smallest attainable p-value is 2^{1-G} rather than 2^{-G}. An exhaustive enumeration at $G = 7$ gives 64 distinct values out of 128 and a minimum p-value of $2/128$ (`verification.json`, entry `E14`). That floor does not bind at any G reported below, the smallest of which is 20.

The second procedure is the restricted wild cluster bootstrap of Cameron et al. (2008), drawing Rademacher weights at the cluster level with (4.2) imposed on the residuals. The implementation partials the unit-by-category and period-by-category effects out once and applies the weights to the cluster residuals of the partialled-out regression, rather than re-absorbing those effects on every draw; the two versions of the statistic differ by $O_p(G^{-1})$. No finite-sample result is claimed for this procedure. Djogbenou et al. (2019) give conditions for asymptotic validity of the wild cluster bootstrap in a linear regression with clustered errors: independence across clusters, moment conditions on the cluster-level scores, and a number of clusters growing without bound with no cluster dominating. Section 9 reports its size in the designs run there

and claims nothing beyond that.

MacKinnon and Webb (2017) report simulations in which the cluster-robust t-test over-rejects more when the $G = 50$ cluster sizes are proportional to United States state populations than when they are balanced, in which the wild cluster bootstrap corrects most of that over-rejection, and in which it fails when only a few clusters are treated. Their designs vary the cluster-size distribution and the number of treated clusters, not the ratio m/G that drives Corollary 2. Every cluster in the simulation of Section 9 holds one unit observed for $T = 120$ periods, so that simulation cannot test their failure mode either.

Implementation

The three procedures are run as follows. Steps 1 and 2 use only the design and can be carried out before any outcome is observed.

1. Form $\tilde{X} = \Pi X$ and $S = \tilde{X}'\tilde{X}$ on the common cell set of Assumption 2, fix the post-treatment block P, and record $m = |P|$. For a contrast $w \in \mathbb{R}^m$ set $a = S^{-1}[\cdot, P]w$ and $\varphi_g = a'S_g a$, and evaluate $G_{\text{eff}}(w)$ from Definition 1. Where one number is wanted for a joint test, take $\min_w G_{\text{eff}}(w)$ over the m coordinate contrasts $w = e_1, \ldots, e_m$, which is the least favorable of the directions the joint statistic combines.

2. Evaluate (5.2) at m and at $G_{\text{eff}}(w)$ in place of G to obtain the size the nominal level-α chi-square test would have, as in Figure 1. If that size is larger than the researcher is willing to carry, either reduce m until it is not, or calibrate by Steps 3 and 4.

3. With $C \geq 5$ categories, use the label-permutation test of Definition 2, enumerating all $C!$ assignments while that is feasible and sampling from S_C otherwise. Proposition 3(iii) rules the two-sided version out at $\alpha = 0.05$ when $C \leq 4$, and Assumption 4 is what buys its finite-sample validity.

4. With $C \leq 4$, or when Assumption 4 is doubtful, use the cluster sign-change test of Theorem 2 or the restricted wild cluster bootstrap, both with the null (4.2) imposed on the residuals and $B - 1$ draws.

5. When the response shape k_P is known up to scale, replace the two-sided Wald statistic by the directed statistic of Theorem 3 inside whichever randomization Step 3 or Step 4 selected. Table 3 reports what that substitution is worth at $G = 51$.

6. To use (5.2) as a reference distribution rather than as a diagnostic, match ω_2 to the finite-sample factor the software applies. Proposition 2 sets $\omega_2 = (n - 2q)/(n - 1)$, counting the $2q$ interacted regressors and not the absorbed unit and period effects, and the factor multiplying the sandwich is then $\{G/(G-1)\}\omega_2^{-1}$. A CR1 implementation that

counts the absorbed effects reports a larger factor, so recover ω_2 by dividing $G/(G-1)$ by the factor the software used. In the test design used to check Proposition 2 the two conventions differ by a factor of 1.178 (`verification.json`, entry `V1b`).

The code implementing Steps 1 to 6 is `src/did_measurement/inference.py` in the replication archive.

8 Power, the minimum detectable response, and the optimal statistic

Let the alternative be $\gamma_\ell = \kappa\, k(\ell)$ for a response path $k(\cdot) \geq 0$ and a scalar magnitude $\kappa > 0$, so that $H_1 : \kappa > 0$. Under local alternatives the noncentrality of the Wald statistic depends on the design through two scalars: $\Sigma_\delta^2 = \sum_c (d_c - \bar{d})^2$, a property of the exposure index, and the timing quantity

$$I_k = k_P' \big([S^{-1}]_{PP} \big)^{-1} k_P, \tag{8.1}$$

where $k_P = (k(\ell))_{\ell \in P}$ is the response shape on the post-treatment block. In the single-cohort design with N units, an adopting share p, T_0 omitted reference periods and the constant path $k \equiv \mathbf{1}$, $[S^{-1}]_{PP} = \{Np(1-p)\}^{-1}(I_m + \mathbf{1}\mathbf{1}'/T_0)$ and $I_k = Np(1-p)\, mT_0/(m+T_0)$, which saturates in T: with a single omitted period the information about a level shift never exceeds $Np(1-p)$, however long the panel. The staggered design of Section 9 gives $I_k = 23.62$.

Corollary 3 (closed-form minimum detectable response)**.** *Let $\lambda^*(\alpha, \beta, m)$ solve $\Pr\{\chi^2_m(\lambda^*) > \chi^2_{m,1-\alpha}\} = 1 - \beta$. With a linear wedge and homoskedastic categories,*

$$\kappa_{\mathrm{MDE}}^{\mathrm{Wald}} = \sigma \sqrt{\frac{\lambda^*(\alpha, \beta, m)}{\Sigma_\delta^2\, I_k}}, \qquad \kappa_{\mathrm{MDE}}^{\mathrm{dir}} = \sigma\, \frac{z_{1-\alpha} + z_\beta}{\sqrt{\Sigma_\delta^2\, I_k}}. \tag{8.2}$$

Both Σ_δ^2 and I_k are computable before any outcome is observed. The third input, σ^2, is the δ^2-weighted mean $\sum_c \delta_c^2 \sigma_c^2 / \sum_c \delta_c^2$ of the category residual variances σ_c^2 of the loading-normalized outcome, and it is the only quantity requiring pilot data; a researcher estimates it by the residual variance of the within-category regression in (4.1) run on the normalized outcome over untreated cells. Every figure below sets $\beta = 0.20$, so power is $1 - \beta = 0.80$, and $\lambda^*(0.05, 0.20, 13) = 17.85$. The expression for $\kappa_{\mathrm{MDE}}^{\mathrm{Wald}}$ is stated against the chi-square critical value, which Proposition 2 shows is the wrong one; it therefore understates the minimum detectable response of any correctly calibrated Wald test. Replacing $\chi^2_{m,1-\alpha}$ by the exact critical value from (5.2) raises the required noncentrality from 17.85 to 41.46 at $G_{\mathrm{eff}} = 36.42$ and $m = 13$, inflating the minimum detectable response by a factor of 1.524 (`results/theory/inference/verification.json`, entry `E11`).

At the design of Section 9 with $C = 5$ and $m = 13$, (8.2) returns $\kappa_{\text{MDE}}^{\text{Wald}} = 0.363$ and $\kappa_{\text{MDE}}^{\text{dir}} = 0.242$ for the linear wedge; solving the same noncentrality equation at the nonlinear wedge that design actually generates raises the Wald figure to 0.450. The simulated magnitude at which size-adjusted power first crosses 0.80 in that design is 0.254 for the directed cluster sign-change test, 0.477 for the directed label-permutation test and 0.566 for the chi-square Wald at its size-adjusted critical value. Those crossings are interpolated on a coarse κ grid with 600 replications per cell and are accurate to roughly 0.05. Sources: `mde_table.csv` for the closed forms, including I_k and $\lambda^*(0.05, 0.20, 13)$ above, and `perm_mde_vs_sim.csv` for the crossings.

The Wald statistic is not the right statistic against this alternative.

Theorem 3 (optimal directed test)**.** *Consider the Gaussian limit experiment in which the category event studies satisfy $b_c \sim N(\tau + \kappa d_c k, \Sigma)$ independently across c, with $\tau \in \mathbb{R}^q$ a free nuisance and Σ common across categories, so that $\hat{\gamma}_P \sim N(\kappa k_P, V)$ exactly with $V = \text{Var}(\hat{\gamma}_P)$, and suppose V and the response shape k_P are known. Among tests of $H_0 : \kappa = 0$ against $H_1 : \kappa > 0$ that are invariant under the group of shifts $b_c \mapsto b_c + a$, $a \in \mathbb{R}^q$, common to all categories, the test rejecting when $t = k_P' V^{-1} \hat{\gamma}_P / \sqrt{k_P' V^{-1} k_P} > z_{1-\alpha}$ is uniformly most powerful, with power $\Phi(\kappa \sqrt{k_P' V^{-1} k_P} - z_{1-\alpha})$.*

Theorem 3 is stated for known V. In practice V is unknown, and the feasible directed statistic substitutes the CR1 estimate built from G clusters; that is the statistic reported in Tables 2 and 3. The substitution costs the normal calibration, and the feasible statistic is the one that the label permutation of Theorem 1 and the cluster sign-change randomization of Theorem 2 calibrate without a normal approximation.

Misspecifying the direction costs power and not level. Running the test at $\tilde{k}$ when the truth is $\gamma_P = \kappa k_P$ makes $t = \tilde{k}' V^{-1} \hat{\gamma}_P / \sqrt{\tilde{k}' V^{-1} \tilde{k}}$ normal with unit variance and mean $\kappa \cos\vartheta \sqrt{k_P' V^{-1} k_P}$, where ϑ is the angle between $\tilde{k}$ and k_P in the inner product $\langle a, b \rangle = a' V^{-1} b$. Its power is $\Phi(\kappa \cos\vartheta \sqrt{k_P' V^{-1} k_P} - z_{1-\alpha})$, so the minimum detectable response is the $\kappa_{\text{MDE}}^{\text{dir}}$ of (8.2) divided by $\cos\vartheta$, and it is unbounded as $\tilde{k}$ approaches orthogonality to k_P in that metric. The test is level α for every choice of $\tilde{k}$.

When k_P is restricted only to the non-negative orthant no uniformly most powerful invariant test exists, and the likelihood-ratio statistic has the chi-bar-square null distribution of Kudô (1963) and Shapiro (1988), a mixture $\sum_{j=0}^{m} w_j(V) \chi_j^2$ in which $w_j(V)$ is the probability that the projection of $N(0, V)$ onto the non-negative orthant in the V^{-1} metric has exactly j positive coordinates. In the designs of Section 9, its size-adjusted power under the label-permutation calibration lies between that of the two-sided Wald and that of the directed statistic at every κ on the grid and at both $C = 5$ and $C = 8$. We have not established that ordering in general.

The ratio of minimum detectable responses depends on how each test is calibrated. Taking

the Wald at the exact critical value from (5.2) and the directed test at the $t_{G_{\text{eff}}-1}$ critical value that its estimated CR1 variance requires, that ratio at power 0.80 is 0.393 at $G_{\text{eff}} = 36.42$, 0.448 at $G_{\text{eff}} = 51$ and 0.516 at $G_{\text{eff}} = 100$ (`results/theory/inference/verification.json`, entry `E11`). Calibrating the directed test at $z_{1-\alpha}$ instead, as Theorem 3 does under known V, gives 0.386, 0.442 and 0.512. At the first of these designs the direction restriction divides the $\Sigma_\delta^2 I_k$ required to detect a given κ, and so the number of clusters required at fixed per-cluster information, by 6.5.

9 Simulations

Design. The simulated panel is a two-layer recording model with count outcomes. A latent factor $F_{it} = f_i + g_t + \tau D_{it} + v_{it}$ with $\tau = 0.4$ drives C category indices $Y^*_{itc} = a_{ic} + \lambda_{tc} + \theta_c F_{it} + \varepsilon_{itc}$, in which the loadings θ_c are a uniformly random permutation of C values evenly spaced in 1 ± 0.3, drawn independently of d. Recording thins counts. With $\mu_{ic} \sim N(0, 0.5^2)$ the untreated recording rate is $r_{ic} = \min\{\max(0.6\, e^{0.2\mu_{ic}}, 0.02), 1\}$; on treated cells it becomes $\tilde{r}_{itc} = \min\{r_{ic}(1 + \kappa k(\ell_{it})d_c), 1\}$ and equals r_{ic} elsewhere; and $Y_{itc} \sim \text{Poisson}(500\, \tilde{r}_{itc}\, e^{0.25 Y^*_{itc}})$. The regression outcome is $\log\{(Y_{itc} + 0.5)/500{,}000\}$, so the recording wedge enters (4.1) as $\log(1 + \kappa k(\ell)d_c)$ rather than as $\kappa k(\ell)d_c$. The response path is flat: $k(\ell) = 1$ at every post-treatment horizon. All disturbances are Gaussian, independent across clusters and serially independent within cluster. The null (4.2) corresponds to $\kappa = 0$.

We vary $G \in \{20, 30, 51, 100\}$ with one unit per cluster, so every cluster carries the same number of observations, and $C \in \{3, 5, 8\}$ with exposure indices equally spaced on $[0.1, 1]$. Every design has $T = 120$, adoption at $t = 48$, 72 and 96 by 20% of units each and 40% never treated, which gives $G_{\text{eff}}/G = 0.714$. Six pre-treatment and twelve post-treatment horizons enter (4.1), omitting $\ell = -1$, so $q = 18$ and $m = 13$ in every cell of every block. Each size cell uses 2,000 replications at nominal level 0.05.

The wild cluster bootstrap uses 499 draws and the cluster sign-change test 999, each including the identity draw, so $\hat{p} = (1 + \#\{b \geq 2 : T_b \geq T_{\text{obs}}\})/B$. The label-permutation test of Definition 2 enumerates all $C!$ assignments at $C = 3$ and $C = 5$. At $C = 8$ it is the Monte Carlo version with $B = 999$, namely 998 sampled assignments together with the identity; the largest level it can attain without exceeding $\alpha = 0.05$ is then $\lfloor \alpha B \rfloor / B = 0.049$, and its resolution floor is $1/B$ rather than the $2/C!$ of Proposition 3. The chi-bar-square statistic uses $B = 199$ at $C = 8$. Monte Carlo standard errors are 0.005 at a rejection rate of 0.05, 0.011 at 0.39 and 0.006 at 0.92. Output files named in this section sit in `results/theory/inference/` unless another directory is given.

Two features of the design bear on what the tables can show. The loadings are drawn independently of d and the ε_{itc} are i.i.d. across categories, so Assumption 4 holds by construction and the permutation rows of Table 2 measure the size of a test to which Theorem 1 applies,

for the null (4.2) under the randomization of Definition 2; the departure from Assumption 4 is priced separately below. Assumption 3 does not hold here: part (i) fails because the cluster scores come from Poisson counts through a nonlinear wedge, and part (ii) fails through the never-treated share. Proposition 2 is therefore an approximation in this design, and the accuracy paragraph below reports the error. Cluster sizes are equal and no disturbance is serially correlated, so nothing in this section tests the cluster-size heterogeneity that MacKinnon and Webb (2017) identify as the bootstrap's failure mode, and nothing tests within-cluster serial dependence.

Size. Table 2 reports null rejection rates for (4.2) at nominal level 0.05. The chi-square calibration rejects at 0.920, 0.693, 0.393 and 0.172 at $G = 20$, 30, 51 and 100 with $C = 3$. Across C at $G = 51$ the rates are 0.393, 0.389 and 0.356 at $C = 3$, 5 and 8, against 0.378, 0.378 and 0.378 predicted by Proposition 2 evaluated at G_{eff}, which moves in C only through ω_2. The measured decline of 0.037 from $C = 3$ to $C = 8$ runs the other way. The two cells are independently seeded, so the standard error of the difference is 0.015 and the decline is 2.5 standard errors of it. One candidate is the loading normalization, whose estimate improves with C, and we have not isolated it.

The second block of rows refers W/m to $F(m, G-1)$, the usual small-sample patch. Proposition 2 puts the correct denominator degrees of freedom at $G - m$, which is 38 rather than 50 at $G = 51$ and $m = 13$, and the patch removes about a quarter of the excess over the nominal level: 0.309 against 0.389 at $G = 51$, $C = 5$. The Hotelling calibration at G, which is the one Proposition 2 licenses under Assumption 3, rejects between 0.082 and 0.175 over the twelve cells. That calibration is model-based rather than randomization-based, and Assumption 3(i) is what it rests on: with centred χ^2_1 cluster contributions, whose skewness is 2.83, its size is 0.184 (`verification_extra.json`, entry `E4`). We report it as a diagnostic and recommend the randomization calibrations of Sections 6 and 7 in its place.

The two-sided label-permutation test is at nominal size for $C \in \{5, 8\}$, ranging from 0.043 to 0.059 over all eight such cells, and its directed version from 0.043 to 0.053. At $C = 3$ both have rejection rate 0.000, which is Proposition 3(iii) rather than a numerical artifact. The wild cluster bootstrap ranges from 0.051 to 0.067. The two-sided cluster sign-change test ranges from 0.022 to 0.066 and is conservative at $G = 20$, where its three rates are 0.027, 0.022 and 0.033; the directed version rejects at 0.058, 0.058 and 0.055 in those same three cells.

The exact null law in the ideal experiment. Assumption 3 is not satisfied by the panel design, so Proposition 2 is checked separately in an experiment built to satisfy it: cluster contributions drawn i.i.d. $N(0, \Sigma)$ with $m = 13$, Σ redrawn for each of three seeds, and 60,000 replications in total at each G. The identity $W = \omega_2 T^2$ holds to a maximum relative error of 4.0×10^{-14} at $G = 20$ and 1.3×10^{-15} at $G = 100$. Referring the statistic to the F law of (5.2) at

the nominal G gives null rejection rates of 0.0495, 0.0489, 0.0502 and 0.0499 at $G = 20$, 30, 51 and 100, quoted to four decimals here because the claim is that they are indistinguishable from the nominal 0.05 rather than merely close to it. Kolmogorov-Smirnov tests of the resulting p-values against the uniform return $p = 0.81$, 0.02, 0.06 and 0.77 at those four G, so the pooled $G = 30$ draws reject uniformity at the 5% level. The closed-form chi-square size implied by (5.2) matches the simulated one to within 0.003 at every G. Source: `anatomy_identity.csv`.

Accuracy of the substitution of G_{eff}. Evaluating (5.2) at G_{eff} from Definition 1 predicts a null rejection rate of 0.378 against the observed 0.389 at $G = 51$, $C = 5$, $m = 13$. Across the twelve (G, C) cells the mean absolute prediction error is 2.37 percentage points using G_{eff} and 13.13 percentage points using G. In this design $G_{\text{eff}} = 36.42$, or 71.4% of G, and it is a design object rather than a random one: its standard deviation across replications is 10^{-13}. The Hotelling calibration evaluated at G_{eff} tracks the nominal level only in the middle of the grid. It rejects at 0.000 in all three cells at $G = 20$, at 0.022, 0.020 and 0.021 at $G = 30$, at 0.047, 0.045 and 0.045 at $G = 51$, and at 0.061, 0.065 and 0.055 at $G = 100$, so the substitution is conservative at twenty clusters and mildly liberal at a hundred.

The design rule. Applying Corollary 2 through (5.2), the largest number of restrictions holding chi-square size below 0.075 is $m = 2$ at $G_{\text{eff}} = 36.4$, $m = 4$ at 72.7, $m = 6$ at 100 and $m = 8$ at 150. Testing $m = 13$ restrictions at $G_{\text{eff}} = 36.4$ gives a predicted size of 0.379. These are evaluations of (5.2) at values of m that no simulated cell runs; every cell of every block in this section has $m = 13$, so the rule has no simulated counterpart here.

When exchangeability fails. A departure from Assumption 4 is priced in a Gaussian limit experiment rather than in the panel: one $N(0, \omega_c)$ draw per category, a single restriction, the same randomization over the $C!$ label assignments, a covariance estimate that treats the categories as homoskedastic, and 20,000 replications. With the inflated variance on the highest-exposure category, the label-permutation test using the Wald statistic rejects at 0.050, 0.064, 0.077, 0.090 and 0.102 at $C = 5$, and at 0.048, 0.062, 0.079, 0.100 and 0.125 at $C = 8$, for variance ratios 1, 2, 4, 9 and 25. Placing the inflated variance on the lowest-exposure category instead moves those rates by at most 0.009. The distortion is monotone in the ratio, and its largest value in the grid stays below the chi-square rate of 0.356 measured at $G = 51$, $C = 8$, $m = 13$. DiCiccio and Romano (2017) show for a regression coefficient that studentizing the statistic makes the permutation test exact under independence and asymptotically valid under heteroskedasticity, so a studentized category-level statistic is the natural repair for this departure; we have not derived its behaviour under Assumption 4.

The sign of the distortion depends on how the variance spread lines up with the exposure index. In the implementation used for Tables 2 and 3, which recomputes the CR1 covari-

ance for every permuted label vector, a $C = 6$ design with category variances decreasing in exposure moves the permutation rejection rate the other way, from 0.047 at variance ratio 1 to 0.021 at ratio 25 over 2,000 replications. Sources: `exchangeability_failure.csv`, and `a2_size_heteroskedastic.csv` under `results/fixes/referee/`.

The price of label randomization. Relative to a normal benchmark at level 0.05 and power 0.80, the label-permutation test requires the standardized signal inflated by a factor of 1.892 at $C = 4$, 1.352 at $C = 5$, 1.235 at $C = 6$, 1.149 at $C = 8$ and 1.098 at $C = 10$, so the price falls as C grows. The factor is a large-sample calculation and it holds up only at the larger C. The minimum detectable response it implies is 0.231 at $C = 8$ against 0.243 in simulation, and 0.327 at $C = 5$ against 0.477, where the simulated value is 46% larger, at the smallest C for which Proposition 3(iii) makes the two-sided test usable. Sources: `perm_power_ceiling.csv` and `perm_mde_vs_sim.csv`.

Power. Table 3 reports size-adjusted power at $G = 51$ against a recording response with a flat path, $k(\ell) = 1$ at every post-treatment horizon, indexed by the magnitude κ, over 600 replications per cell. Two patterns run in opposite directions. Label randomization with the two-sided Wald statistic is the least powerful procedure at every κ at or above 0.10 in both C of Table 3: at $C = 5$ and $\kappa = 0.45$ it attains 0.383 against 0.705 for the Wald statistic at its oracle critical value. At $\kappa = 0.05$ the three two-sided procedures sit within one Monte Carlo standard error of one another, at 0.083, 0.082 and 0.080 at $C = 5$, so the ordering is not measured there. Directing the statistic more than reverses the loss. The directed cluster sign-change test attains 0.983 at $C = 5$ and $\kappa = 0.45$, and at $C = 8$ it reaches 0.842 at $\kappa = 0.20$ where the two-sided Wald reaches 0.277. The gain comes from the direction restriction rather than from the randomization.

The directed statistics in Table 3 use the contrast $w = \mathbf{1}_m$, which is the true response path in every cell of the block, since $k(\ell) = 1$ throughout. Theorem 3 is stated for known k_P, and the table measures the directed tests under that condition. No cell was run in which the assumed direction differs from the true one, so these entries bound what a researcher who has to guess the shape can expect, and the table carries no evidence on how much of the gain survives a wrong guess.

The size-adjusted power of the two-sided permutation test is non-monotone in κ at $C = 5$, falling from 0.452 at $\kappa = 0.60$ to 0.383 at $\kappa = 0.80$. The two rates come from independently seeded replications, so the standard error of the difference is 0.028 and the decline of 0.068 is 2.4 of them.

The same disagreement appears twice more at $C = 5$. The large-sample local power calculation for the label-permutation test predicts 0.538, 0.768 and 0.927 at $\kappa = 0.45$, 0.60 and 0.80 against simulated 0.383, 0.452 and 0.383, overstatements of 15, 32 and 54 percentage

points, and it predicts 0.041 at $\kappa = 0$ against the nominal level of 0.05. The minimum detectable response it implies is 0.327 against a simulated 0.477, while at $C = 8$ the same calculation gives 0.231 against 0.243. The non-monotonicity and the theory-simulation gap are the same disagreement at small C, and we have not isolated its cause. Sources: `local_power_check.csv` and `perm_mde_vs_sim.csv`.

Exposure spread, panel length and the pre-window. Three further blocks vary the design rather than the procedure, at $G = 51$ and $C = 5$. Compressing the exposure indices from $[0.1, 1]$ to $[0.4, 0.6]$ cuts Σ_δ^2 from 0.506 to 0.025 and raises the closed-form minimum detectable response of the Wald test from 0.363 to 1.639, a factor of 4.5, with the directed test moving from 0.242 to 1.854. Halving the panel to $T = 60$ moves the Wald figure by under 2%, from 0.363 to 0.370. Replacing the six-period pre-window by the single omitted period $\ell = -1$ raises I_k from 23.6 to 448 and cuts the closed-form directed minimum detectable response to 0.050, against 0.081 for the simulated directed cluster sign-change test; the chi-square null rejection rate in that block is 0.405 over 600 replications, so narrowing the pre-window buys power without repairing the calibration. Sources: `mde_table.csv` and `reference_block_table.csv`.

10 Application

We apply the procedures to two state panels assembled for a companion project on treatment-induced recording change. The identification claim in each case is that under measurement invariance the post-treatment event-study coefficients do not vary with the exposure index, so (4.2) holds. The application is illustrative of the inference problem rather than a contribution to either substantive literature.

The first panel is a state-by-year-by-offense panel of recorded index crime covering 51 states from 1979 to 2024, clustered on state. Recorded counts are the FBI's summary-based estimated state totals and the outcome is the log offense rate per 100,000 residents. The treatment is a state's transition from summary reporting to the National Incident-Based Reporting System, dated as the first year in which at least half of the state's agencies on the FBI Crime Data Explorer roster carry a NIBRS start date; Florida, Hawaii, New York and Pennsylvania do not reach that share by 2024 and enter as never-treated. Six offense categories carry the gradient, in ascending order of exposure: motor-vehicle theft, aggravated assault, robbery, burglary, rape and larceny. Homicide is retained in the panel as a $d = 0$ series and is not used in the gradient. The exposure index is $d_c = 1 - r_c$, with r_c the share of victimizations reported to police averaged over the 2018 and 2019 National Crime Victimization Survey (Bureau of Justice Statistics, *Criminal Victimization 2019*, NCJ 255113, Table 6) and rescaled so the largest focal value is one, giving $d = (0.290, 0.604, 0.628, 0.699, 0.976, 1.000)$ in that order. The design has $G = 51$, $C = 6$, $m = 9$ and $\Sigma_\delta^2 = 0.349$, and the multiset $\{\delta_c\}$ is not symmetric

Table 2: Null rejection rates, nominal level 0.05

		Clusters G			
Statistic and calibration	C	20	30	51	100
Wald, χ^2_m	3	0.920	0.693	0.393	0.172
	5	0.905	0.686	0.389	0.187
	8	0.917	0.690	0.356	0.172
W/m, $F(m, G-1)$	3	0.838	0.586	0.311	0.143
	5	0.821	0.558	0.309	0.153
	8	0.834	0.571	0.283	0.138
Hotelling at G	3	0.175	0.164	0.122	0.082
	5	0.174	0.160	0.124	0.092
	8	0.172	0.167	0.106	0.088
Hotelling at G_{eff}	3	0.000	0.022	0.047	0.061
	5	0.000	0.020	0.045	0.065
	8	0.000	0.021	0.045	0.055
Label permutation, Wald	3	0.000	0.000	0.000	0.000
	5	0.053	0.050	0.046	0.059
	8	0.051	0.051	0.043	0.058
Label permutation, directed	3	0.000	0.000	0.000	0.000
	5	0.052	0.053	0.048	0.052
	8	0.046	0.050	0.043	0.051
Wild cluster bootstrap, Wald	3	0.053	0.055	0.053	0.061
	5	0.064	0.062	0.051	0.067
	8	0.061	0.061	0.051	0.056
Cluster sign-change, Wald	3	0.027	0.036	0.046	0.058
	5	0.022	0.041	0.050	0.066
	8	0.033	0.041	0.048	0.053
Cluster sign-change, directed	3	0.058	0.056	0.044	0.045
	5	0.058	0.061	0.057	0.046
	8	0.055	0.055	0.057	0.065

Note: rejection rates for the joint null $H_0 : \gamma_P = 0$ of (4.2) in the pooled gradient regression (4.1), estimated by OLS with the CR1 covariance clustered on g, at nominal level 0.05. Design: $m = 13$ post-treatment restrictions, $q = 18$ included horizons, $T = 120$, adoption at $t = 48$, 72 and 96 by 20% of units each with 40% never treated, one unit per cluster, exposure indices equally spaced on $[0.1, 1]$. The Hotelling rows refer W to the F law of (5.2) evaluated at G and at $G_{\text{eff}} = 0.714\,G$. Label permutation randomizes the $C!$ assignments of the exposure index to the category data arrays, enumerating all of them at $C = 3$ and $C = 5$ and using 998 sampled assignments together with the identity at $C = 8$. The wild cluster bootstrap uses 499 draws and the cluster sign-change test 999, each including the identity draw. The directed rows use the contrast $w = \mathbf{1}_m$. 2,000 replications per cell; the Monte Carlo standard error is 0.005 at a rate of 0.05, 0.011 at 0.39 and 0.006 at 0.92. Sources: `results/theory/inference/size_table.csv` and `anatomy_table.csv`.

Table 3: Size-adjusted power against a flat recording response of magnitude κ, $G = 51$

		Recording response κ							
Statistic and calibration	C	0.05	0.10	0.15	0.20	0.30	0.45	0.60	0.80
Wald, oracle critical value	5	0.082	0.107	0.145	0.212	0.362	0.705	0.828	0.872
	8	0.058	0.113	0.150	0.277	0.537	0.850	0.930	0.953
Wald, wild cluster bootstrap	5	0.080	0.110	0.147	0.220	0.362	0.715	0.830	0.872
	8	0.053	0.112	0.157	0.273	0.532	0.852	0.925	0.950
Wald, label permutation	5	0.083	0.103	0.138	0.170	0.212	0.383	0.452	0.383
	8	0.055	0.097	0.142	0.187	0.395	0.655	0.733	0.722
Directed, label permutation	5	0.147	0.250	0.403	0.518	0.657	0.797	0.815	0.772
	8	0.175	0.370	0.535	0.732	0.890	0.973	0.978	0.970
Directed, cluster sign-change	5	0.143	0.320	0.527	0.688	0.895	0.983	0.987	1.000
	8	0.155	0.403	0.613	0.842	0.968	0.997	0.997	1.000

Note: size-adjusted rejection rates against the alternative $\gamma_\ell = \kappa k(\ell)$ with $k(\ell) = 1$ at every post-treatment horizon, for the joint null $H_0 : \gamma_P = 0$ of (4.2). Each entry uses the critical value set at the empirical 5% quantile of that procedure's p-value distribution under $\kappa = 0$ in the same design. Because the randomization p-values have atoms, that rule does not deliver a null rate of 0.05 in every row. At $C = 5$ the achieved rates are 0.050 for the oracle Wald and the directed sign-change rows, 0.051 for the bootstrap, 0.056 for the directed permutation row and 0.060 for the two-sided permutation row, so the two permutation rows at $C = 5$ run at a higher level than the rows beside them; at $C = 8$ every row is between 0.050 and 0.051. The first row is not a feasible test: the chi-square calibration has null rejection rate 0.389 at $C = 5$, and its size-adjusted cut is a nominal p-value of 7.3×10^{-5}, which cannot be computed without the null simulation. It is reported as a power benchmark. For the other rows the adjustment moves the cut only to between 0.043 and 0.067 in p-value units. The directed rows use the contrast $w = \mathbf{1}_m$, which is the true response path in this block. The block also ran $C = 3$, where the only attainable two-sided permutation p-values are 1/3, 2/3 and 1 and the adjusted rule reduces to rejecting at $\hat{p} = 1/3$, with an achieved null rate of 0.340; those cells are omitted. Design: $G = 51$ clusters, $m = 13$ post-treatment restrictions, $q = 18$ horizons, $T = 120$, adoption at $t = 48$, 72 and 96, exposure indices equally spaced on $[0.1, 1]$. Label permutation enumerates all $C!$ assignments at $C = 5$ and uses 998 sampled assignments together with the identity at $C = 8$; the cluster sign-change test uses 999 draws including the identity. 600 replications per cell; Monte Carlo standard error at a rate of 0.5 is 0.020. Sources: `results/theory/inference/power_curves.csv` and `size_adjustment_audit.csv`.

about zero.

Each category array is divided by an estimated loading before the gradient is formed, because the categories respond by different amounts to the common latent component, and the choice of normalization determines whether Theorem 1 applies. We report three. The naive normalization sets every loading to one. The triad normalization is the log-triad estimator (6.1) of Section 6, computed from the untreated cross-category covariance matrix and scaled to geometric mean one. Relabeling the categories relabels $\hat{\rho}$ identically, so by Lemma 3 the normalized arrays are an S_C-equivariant function of the raw arrays, the Wald statistic is order-invariant, and Theorem 1 applies. The third normalization is the one the specification was originally run with, which fixes its reference category at the position of the smallest d_c. That choice makes the statistic depend on the order in which the pairs $(\delta_c, \mathcal{Y}_c)$ are listed, so Theorem 1 does not cover it, and we report it for comparison only.

Under the triad normalization the chi-square test of (4.2) rejects and none of the three randomization calibrations does. The design has $G_{\text{eff}} = 45.9$, which by the rule of Section 9 supports $m = 3$ restrictions at size 0.075; the specification tests $m = 9$, for which (5.2) predicts a chi-square size of 0.169. The chi-square statistic is 19.00 with $p = 0.025$. The label-permutation test of Definition 2 over all 720 assignments of δ returns $p = 0.115$, the restricted wild cluster bootstrap returns $p = 0.092$ and the cluster sign-change test returns $p = 0.104$. Referring W to (5.2) with G_{eff} in place of G returns $p = 0.115$; that substitution is the approximation of Definition 1 and not a calibration Proposition 2 licenses, and Section 9 prices its null rejection rate at 0.000 at $G = 20$ rising to 0.061 at $G = 100$, so it is reported here as a diagnostic. The realized resolution floor of the permutation test here is $1/720 = 0.0014$, so rejection at the 5% level was attainable and the non-rejection is not a consequence of Proposition 3.

The chi-square rejection is joint with the normalization. Under the naive normalization the chi-square test does not reject at all, at $p = 0.141$, and the label permutation returns $p = 0.396$; under the reference-category normalization the chi-square test gives $p = 0.018$ and the label permutation $p = 0.096$. Table 4 reports all three normalizations for each of the nine crime designs, of which the design described above is one. The artifact that correct calibration overturns is therefore an artifact of the chi-square reference distribution applied to a normalized outcome; under the naive normalization there is nothing to overturn.

A pre-period placebo in the same design behaves differently. Under the triad normalization its chi-square p-value is 6×10^{-6} and its label-permutation p-value over the same 720 assignments is 0.0167, the twelfth largest of the 720 orbit values. The exposure indices are not symmetric about zero and the realized orbit multiplicity is one, so Proposition 3(ii) does not apply and the floor is $1/720$ rather than $2/720$. Under the naive normalization the same permutation test returns $p = 0.153$, so the pre-period evidence, like the post-period evidence, is conditional on the normalization.

The second panel is a state-by-year-by-condition panel of BRFSS crude prevalence covering 51 jurisdictions from 2011 to 2024, clustered on state. The treatment is a state's adoption of the ACA Medicaid expansion, with the cohort set to the calendar year of the coverage effective date when coverage began on or before July 1 and to the following year otherwise; ten states do not expand through 2024. Seven conditions carry the gradient, and the exposure index grades them by whether a clinical encounter is required for the respondent to report the condition: diagnosed diabetes 1.00, diagnosed high cholesterol 1.00, diagnosed hypertension 0.90, diagnosed depression 0.80, self-reported obesity 0.05, current smoking 0.00 and no exercise in the past month 0.00. Three further series in the panel, diabetes mortality, insurance coverage and a past-year checkup, are not used in the gradient.

Hypertension and high cholesterol are asked only in odd years, so Assumption 2 fails on the seven-category stack. That failure costs the closed form of Lemma 2 and not the conclusion of

Theorem 1, which needs Assumption 4 in its place. What blocks exactness on the full set is that the cell pattern is a function of the category label, so Assumption 4 fails as well. The cluster-level procedures of Section 7 use neither assumption and could be run on the unbalanced full set; we do not report that specification. We report two balanced sub-designs: the five annually asked categories on all 709 state-year cells, and all seven categories on the 353 odd-year cells. Under either equivariant normalization no category is dropped, so $C = 5$ and $C = 7$, with $G_{\text{eff}} = 36.0$ and 32.8, $\Sigma^2_\delta = 0.958$ and 1.444, and support for $m = 2$ restrictions against the $m = 7$ tested. Neither sub-design rejects (4.2) under the triad normalization, at chi-square $p = 0.730$ and 0.391 and label-permutation $p = 0.850$ and 0.854 over all $C!$ assignments, nor under the naive normalization, at chi-square 0.295 and 0.967 and label permutation 0.867 and 0.964.

The reference-category normalization gives a different answer, and it is the one Theorem 1 does not cover. Its loading guard drops any category whose estimated loading is below 0.1 in absolute value: diagnosed diabetes, the $d = 1$ series the design is built around, in the annual sub-design, leaving $C = 4$; high cholesterol and no exercise in the odd-year sub-design, leaving $C = 5$. In the annual sub-design the chi-square test returns $p = 0.100$, the Hotelling calibration at G_{eff} returns 0.230, the label permutation 0.333 and the wild cluster bootstrap 0.261, while the directed cluster sign-change test rejects at $p = 0.042$. At $C = 4$ the realized resolution floor of the two-sided label-permutation test is $2/24 = 0.083$, so by Proposition 3 its non-rejection carries no information at the 5% level and only the cluster-level procedures of Section 7 are usable there. In the odd-year sub-design the chi-square test returns $p = 0.443$ while the two-sided label permutation rejects at $p = 0.033$, the reverse of the crime pattern.

Where the guard fires, permuting only the surviving labels conditions on a multiset that is not ancillary. Under Assumption 4 the reference set for the C surviving arrays is the set of $C_0!/(C_0 - C)!$ ordered injections of those arrays into the full C_0-vector of labels, not the $C!$ permutations of the survivors. Recomputing against the injection set moves the annual sub-design from 0.333 to 0.317 over 120 injections, with realized floor $2/120 = 0.017$, and the odd-year sub-design from 0.033 to 0.062 over 2,520 injections. The odd-year rejection is an artifact of the reference set, and neither number is exact for (4.2) in the sense of Theorem 1, because the normalization that made the guard fire is not S_C-equivariant.

Under the log outcome the odd-year sub-design rejects under every calibration when the reference-category normalization is used: chi-square 3×10^{-6}, Hotelling at G_{eff} 0.0024, label permutation 0.017, restricted wild cluster bootstrap 0.021 and cluster sign-change 0.022. The guard does not fire there, so the injection and label reference sets coincide. Under the triad and naive normalizations of the same log outcome nothing rejects, at chi-square 0.213 and 0.960 and label permutation 0.485 and 0.965.

Neither non-rejection is evidence that measurement invariance holds, because in both panels the minimum detectable response from (8.2) exceeds the magnitude the substantive mechanism

implies, by a factor of 3.4 in the crime panel and 17 in the health panel. Evaluating (8.2) at $\alpha = 0.05$, $\beta = 0.20$ and each design's own Σ_δ^2 and I_k, with σ^2 set to the median category residual variance, gives $\kappa_{\text{MDE}}^{\text{Wald}} = 0.255$ log points per unit d for the crime design at $m = 9$, $\Sigma_\delta^2 = 0.349$ and $I_k = 40.5$ under the naive normalization, and 0.326 under the reference-category normalization. The recording wedges that the companion project's own bounds identify as the magnitudes that would matter run from 0.008 to 0.074 log points per unit d. For the health sub-designs the same expression gives 0.863 and 0.930 percentage points of prevalence per unit d at $m = 7$, $\Sigma_\delta^2 = 0.958$ and 1.444 and $I_k = 22.4$ and 13.6, against a mechanism-implied response of 0.052 percentage points for diagnosed diabetes at the estimated 1.98 percentage point coverage gain and an undiagnosed share of 0.213. These values are stated against the chi-square critical value, which Section 8 shows understates the detectable response of a correctly calibrated test.

Data availability. The recorded-crime series are the FBI's summary-based estimated state totals; the NIBRS adoption dates come from the FBI Crime Data Explorer agency API; the reporting rates defining d_c come from Bureau of Justice Statistics, *Criminal Victimization 2019*, NCJ 255113, Table 6. The health series are BRFSS crude prevalence from the CDC Socrata dataset `dttw-5yxu`, and the expansion effective dates come from the Kaiser Family Foundation table of state Medicaid expansion decisions. All are public. The two constructed panels and the code that builds them are in the replication archive named in the title footnote. Every p-value, G_{eff} and predicted size reported in this section is read from `results/fixes/empirics/inference_gradient.csv`, and every realized resolution floor from `results/fixes/empirics/app_floors_injection.csv`.

11 Conclusion

The chi-square calibration of a cross-outcome falsification test is oversized at the cluster counts these tests are run on: at $G = 51$ and $m = 13$ its size is 0.389 against a nominal 0.05. Under equal cluster design weights the statistic is a scaled Hotelling T^2, giving the rate $m = o(G^{2/3})$ at which chi-square stays valid. The label-permutation test of Definition 2 is exact in finite samples for (4.2) under Assumption 4, with no condition on G; the cluster sign-change test is exact under sign symmetry of the cluster contributions, which the feasible statistic attains at every G under Assumption 3(ii); the bootstrap carries no exactness claim. Their simulated sizes in Table 2 are 0.043–0.059, 0.022–0.066 and 0.051–0.067 at $C \in \{5, 8\}$. In the crime design the chi-square test rejects at $p = 0.025$ and none of the three does.

Two limitations bound what is claimed. Proposition 2 needs equal design weights across clusters, which no design here satisfies; where that fails its F calibration is neither exact nor conservative, so the result explains the distortion rather than replacing it. The second is

Table 4: Crime application: p-values for $H_0 : \gamma_P = 0$ across nine designs and three loading normalizations

Design	Normalization	C	G_{eff}	χ^2	Hotelling	Perm.	Bootstrap	Sign
Baseline	adjusted	6	45.9	0.018	0.097	0.096	0.075	0.083
	naive			0.141	0.304	0.396	0.277	0.311
	triad			0.025	0.115	0.115	0.092	0.104
Pre-2020	adjusted	6	33.2	0.367	0.604	0.782	0.630	0.593
	naive			0.596	0.769	0.911	0.773	0.766
	triad			0.422	0.647	0.813	0.670	0.639
Drop rape	adjusted	5	45.9	0.016	0.089	0.075	0.063	0.048
	naive			0.008	0.062	0.067	0.040	0.052
	triad			0.018	0.096	0.083	0.066	0.051
Drop rape, larceny	adjusted	4	45.9	0.107	0.258	0.667	0.218	0.209
	naive			0.139	0.300	0.667	0.254	0.303
	triad			0.106	0.257	0.667	0.220	0.203
Cohort +1	adjusted	6	45.7	0.017	0.094	0.117	0.079	0.078
	naive			0.120	0.277	0.371	0.258	0.299
	triad			0.023	0.110	0.122	0.097	0.099
Threshold 0.8	adjusted	6	38.0	0.182	0.394	0.243	0.411	0.367
	naive			0.000	0.007	0.011	0.002	0.003
	triad			0.204	0.419	0.268	0.431	0.399
Core agency types	adjusted	6	46.9	0.012	0.074	0.032	0.062	0.062
	naive			0.164	0.328	0.389	0.306	0.359
	triad			0.018	0.093	0.051	0.083	0.091
Drop never-treated	adjusted	6	45.3	0.017	0.094	0.172	0.093	0.094
	naive			0.079	0.220	0.338	0.216	0.245
	triad			0.026	0.119	0.207	0.121	0.131
Core types, pre-2020	adjusted	6	35.8	0.320	0.547	0.793	0.566	0.554
	naive			0.537	0.717	0.926	0.728	0.717
	triad			0.367	0.587	0.819	0.601	0.602

Note: p-values for the joint null $H_0 : \gamma_P = 0$ of (4.2) in the pooled gradient regression (4.1), estimated by OLS with the CR1 covariance clustered on state. Every design has $G = 51$ state clusters except drop never-treated, which has $G = 47$, and every design tests $m = 9$ post-treatment restrictions. χ^2 is the χ^2_9 calibration; Hotelling is (5.2) evaluated at G_{eff}; Perm. is the label-permutation test of Definition 2 enumerating all $C!$ assignments of δ, whose realized resolution floor is 1/720 at $C = 6$, 1/120 at $C = 5$ and 1/24 at $C = 4$; Bootstrap is the restricted wild cluster bootstrap and Sign the cluster sign-change test, each with 1,999 draws. Naive sets every loading to one; triad is the S_C-equivariant estimator (6.1); adjusted fixes the reference category at the smallest d_c and is not covered by Theorem 1. The entry shown as 0.000 is 2.1×10^{-5}. Sources: `results/fixes/empirics/inference_gradient.csv` for the p-values and `results/fixes/empirics/app_floors_injection.csv` for the realized floors.

Assumption 4, which fails when categories have different error variances. At a variance ratio of 25 the simulated size of the label-permutation test is 0.102 at $C = 5$ and 0.125 at $C = 8$. The cluster-level procedures do not use it.

Lemma 2 requires a category-balanced design, which the health panel shows binding, and what blocks Theorem 1 there is Assumption 4, because the cell pattern depends on the category label. Extending the reduction to unbalanced panels would recover the procedure where categories are surveyed on different schedules.

A Proofs

Proof of Lemma 2. Under Assumption 2 the regressor block in (4.1) is common across categories after the two-way within transformation, so stacking the C categories gives design matrix $[\,\mathbf{1}_C \otimes \tilde{X},\ \delta \otimes \tilde{X}\,]$ with $\delta = (\delta_1, \ldots, \delta_C)'$. Because $\mathbf{1}_C'\delta = 0$ by the centering of δ, the two blocks are orthogonal, and the normal equations separate into $(\mathbf{1}_C'\mathbf{1}_C)S\hat{\beta} = \sum_c \tilde{X}'\tilde{y}_c$ and $(\delta'\delta)S\hat{\gamma} = \sum_c \delta_c \tilde{X}'\tilde{y}_c$. Dividing by C and $\sum_c \delta_c^2$ respectively and using $b_c = S^{-1}\tilde{X}'\tilde{y}_c$ gives the stated expressions. □

Proof of Proposition 2. By (5.1) the CR1 covariance of $\hat{\gamma}$ on the post block is $\hat{V}_{PP} = \omega \sum_g \eta_{gP}^f \eta_{gP}^{f\prime}$ with $\omega = \omega_2^{-1} G/(G-1)$. The within-category residual is $e_{gc} = u_{gc} - \tilde{X}_g S^{-1}\tilde{X}'u_c$ by Lemma 2, so $a_g^f = a_g - S_g S^{-1}\sum_h a_h$ and $\eta_g^f = \eta_g - S^{-1}S_g \sum_h \eta_h$. Under Assumption 3(ii), $S^{-1}S_g = I_q/G$, so $\eta_g^f = \eta_g - \bar{\eta}$, the restriction to P commutes with the correction, and $\sum_g \eta_{gP}^f \eta_{gP}^{f\prime} = (G-1)\hat{S}_P$ while $\hat{\gamma}_P - \gamma_P = G\bar{\eta}_P$. Substituting,

$$W = \hat{\gamma}_P'\hat{V}_{PP}^{-1}\hat{\gamma}_P = \omega_2\, \bar{\eta}_P' \Big[\tfrac{1}{G-1}\textstyle\sum_g (\eta_{gP} - \bar{\eta}_P)(\eta_{gP} - \bar{\eta}_P)'\Big]^{-1} \bar{\eta}_P\, G = \omega_2 T^2,$$

where the centering is the within-category fitting correction and not a consequence of imposing (4.2). This is an algebraic identity and does not use Assumption 3(i).

For the distribution, Assumption 3(i) gives

$$\sqrt{G}\,\bar{\eta}_P \sim N(0, \Sigma_{PP}), \qquad (G-1)\hat{S}_P \sim W_m(\Sigma_{PP}, G-1),$$

independently, by Cochran's theorem applied to the i.i.d. Gaussian sample $\eta_{1P}, \ldots, \eta_{GP}$. Hence T^2 is Hotelling's T^2 with dimension m and $G-1$ degrees of freedom (Hotelling 1931), and the standard transformation $\frac{G-m}{(G-1)m}T^2 \sim F(m, G-m)$ holds for $G - m \geq 1$. Substituting $T^2 = W/\omega_2$ gives (5.2).

For the last claim, write $\sum_g \eta_{gP}^f \eta_{gP}^{f\prime} = E'E$ with E the $G \times m$ matrix whose gth row is $\eta_{gP}^{f\prime}$. Summing $\eta_g^f = \eta_g - S^{-1}S_g \sum_h \eta_h$ over g and using $\sum_g S_g = S$ gives $\sum_g \eta_g^f = 0$, so the columns of E are orthogonal to $\mathbf{1}_G$ and $\operatorname{rank} E \leq \min(m, G-1)$. If $G - 1 < m$ the meat is singular, $\hat{S}_P$ is not invertible, and the Hotelling representation is unavailable. □

Proof of Corollary 2. By (5.2) the exact size of the nominal chi-square test is $\Pr\{F(m, G-m) > (G-m)c_m/(\omega_2(G-1)m)\}$ with $c_m = \chi^2_{m,1-\alpha}$. Write $F(m, G-m) \stackrel{d}{=} (\chi^2_m/m)/(\chi^2_{G-m}/(G-m))$ with the two chi-squares independent, so the size equals $\Pr\{\chi^2_m > c_m\, \omega_2^{-1}(G-1)^{-1}\chi^2_{G-m}\}$.

For (i), write $W \stackrel{d}{=} R\,\chi^2_m$ with $R = \omega_2(G-1)/\chi^2_{G-m}$ independent of χ^2_m, so the size is $\mathbb{E}[h(R)]$ with $h(x) = \Pr\{\chi^2_m > c_m/x\}$ and $h(1) = \alpha$. Put $\nu = G-1$ and $d = G-m$. At fixed m, $\mathbb{E}[R-1] = (m+1)/\nu - (1-\omega_2) + O(\nu^{-2})$ and $\mathbb{E}[(R-1)^2] = 2/\nu + O(\nu^{-2})$, using $\mathbb{E}[\chi_d^{-2k}] = \prod_{j=1}^{k}(d-2j)^{-1}$ for $d > 2k$ and $G(1-\omega_2) = O(1)$. The same formula gives the third central moment of $1/\chi^2_d$ as $16/\{(d-2)^3(d-4)(d-6)\}$ for $d > 6$, whence the signed third moment $\mathbb{E}[(R-1)^3] = O(\nu^{-2})$.

A second-order expansion does not deliver the remainder. Bounding it by $\sup|h'''|\,\mathbb{E}|R-1|^3$ gives $O(\nu^{-3/2})$, because the absolute third moment, unlike the signed one, gains nothing from cancellation. Expand instead to third order with a fourth-order Lagrange remainder. Outside $|R-1| \le 1/2$ the contribution to $\mathbb{E}[h(R)]$ is exponentially small in ν, since h is bounded by one and the chi-square Chernoff bound gives $\Pr(|R-1| > 1/2) \le e^{-c\nu}$ for some $c > 0$. Inside, $\sup_{|x-1|\le 1/2}|h''''(x)|$ is finite and $\mathbb{E}[(R-1)^4] = 3\{\mathbb{E}[(R-1)^2]\}^2 + O(\nu^{-3}) = 12/\nu^2 + O(\nu^{-3})$, so the third-order term and the remainder are both $O(\nu^{-2})$.

Substituting the moments with $h'(1) = c_m f_m(c_m)$ and $h''(1) = -c_m^2 f'_m(c_m) - 2c_m f_m(c_m)$, and using $f'_m(x) = f_m(x)\{(m/2-1)/x - 1/2\}$ so that $c_m^2 f'_m(c_m) = c_m f_m(c_m)\{(m/2-1) - c_m/2\}$, the bracket collapses to

$$h'(1)\,\mathbb{E}[R-1] + \tfrac{1}{2}h''(1)\,\mathbb{E}[(R-1)^2] = c_m f_m(c_m)\Big[\frac{m+c_m}{2\nu} - (1-\omega_2)\Big],$$

which is the stated expansion. At the design of Section 9 with $G = 51$ and $C = 3$, where $\omega_2 = 0.9981$, the two terms in that bracket are 0.354 and 1.9×10^{-3} at $m = 13$ and $\alpha = 0.05$, and $G(1-\omega_2)$ is at most 0.098 over the twelve designs of Table 2 (`results/theory/inference/anatomy_table.csv`).

For (ii), take logarithms. The event is $\log\chi^2_m - \log\chi^2_{G-m} > \log c_m - \log(\omega_2(G-1))$. With $m \to \infty$, $\log\chi^2_m$ is asymptotically normal with mean $\log m - 1/m + O(m^{-2})$ and variance $2/m$, and similarly for $\log\chi^2_{G-m}$ with $G-m \to \infty$. Standardizing and using $\log c_m = \log m + \sqrt{2/m}\,z_{1-\alpha} + o(m^{-1/2})$ gives size $\to \Phi(\chi - z_{1-\alpha})$ with $\chi = \lim\sqrt{m/2}\,\log\{(G-1)/(G-m)\}$. Expanding $\log\{(G-1)/(G-m)\} = (m-1)/(G-1) + O((m/G)^2)$ gives $\chi_{m,G} = m^{3/2}/(\sqrt{2}G) + o(1)$ when $m^{5/2}/G^2 \to 0$. If $m/G \to r \in (0,1)$ then $\log\{(G-1)/(G-m)\} \to -\log(1-r) > 0$ and $\chi_{m,G} \to \infty$, so the size tends to one. □

Proof of Lemma 3. Under Assumption 2 the two-way within transformation is the same operator for every category, so relabelling the arrays gives $\hat{\Sigma}(\mathcal{D}_\sigma)_{ck} = \hat{\Sigma}_{\sigma(c)\sigma(k)}$ and hence $L(\mathcal{D}_\sigma)_{ck} = L_{\sigma(c)\sigma(k)}$. Substituting $j = \sigma(k)$ maps $\{k \ne c\}$ bijectively onto $\{j \ne \sigma(c)\}$, and substituting $(j,l) = (\sigma(k), \sigma(l))$ maps $\{k \ne l,\ k, l \ne c\}$ bijectively onto $\{j \ne l,\ j, l \ne \sigma(c)\}$, so both averages in (6.1) are evaluated at the relabelled index and $\hat{a}_c(\mathcal{D}_\sigma) = \hat{a}_{\sigma(c)}(\mathcal{D})$. The

geometric-mean scaling divides by $\exp\{C^{-1}\sum_j \hat{a}_j\}$, which is σ-invariant, so $\hat{\rho}_c(\mathcal{D}_\sigma) = \hat{\rho}_{\sigma(c)}(\mathcal{D})$. The retained set and the normalized arrays are therefore relabelled with the loadings. □

Proof of Theorem 1. The argument is the classical group-invariance construction; see Lehmann and Romano (2005, Ch. 15). Order-invariance of $\mathcal{T}$ gives, for every $\pi \in S_C$, $\mathcal{T}(\mathcal{D}, \delta_\pi) = \mathcal{T}(\mathcal{D}_{\pi^{-1}}, \delta)$ where $\mathcal{D}_{\pi^{-1}} = (\mathcal{Y}_{\pi^{-1}(1)}, \ldots, \mathcal{Y}_{\pi^{-1}(C)})$: the statistic is computed from the multiset $\{(\delta_{\pi(c)}, \mathcal{Y}_c)\}_c$, and re-indexing by $j = \pi(c)$ turns this into $\{(\delta_j, \mathcal{Y}_{\pi^{-1}(j)})\}_j$.

Under Assumption 4, $\mathcal{D}_{\pi^{-1}} \overset{d}{=} \mathcal{D}$ conditional on $\mathcal{Z}$ for each fixed π. Since S_C is a finite group acting on the index set, the vector $(T(\pi))_{\pi \in S_C}$ is exchangeable conditional on $\mathcal{Z}$: for any $\sigma \in S_C$, $(T(\sigma\pi))_\pi \overset{d}{=} (T(\pi))_\pi$.

We bound $\Pr(\hat{p} \le \alpha)$ for the $\hat{p}$ of Definition 2 directly, which covers ties. For $\sigma \in S_C$ put $N(\sigma) = \#\{\pi' \in S_C : T(\pi') \ge T(\sigma)\}$ and $\mathcal{A} = \{\sigma : N(\sigma) \le \alpha M\}$, so the test rejects exactly on the event $\mathrm{id} \in \mathcal{A}$. If $\mathcal{A}$ is not empty, let σ^* attain the smallest value of T on it. Every $\sigma \in \mathcal{A}$ has $T(\sigma) \ge T(\sigma^*)$, so $\mathcal{A} \subseteq \{\pi' : T(\pi') \ge T(\sigma^*)\}$ and $|\mathcal{A}| \le N(\sigma^*) \le \alpha M$, hence $|\mathcal{A}| \le \lfloor \alpha M \rfloor$ because $|\mathcal{A}|$ is an integer. Write $\phi(t) = \mathbf{1}\{\#\{\pi : t_\pi \ge t_{\mathrm{id}}\} \le \alpha M\}$ for a vector t indexed by S_C, so that the rejection indicator is $\phi\big((T(\pi))_\pi\big)$. Because $\pi \mapsto \sigma\pi$ is a bijection, $\phi\big((T(\sigma\pi))_\pi\big) = \mathbf{1}\{N(\sigma) \le \alpha M\}$, and summing over σ gives $\sum_{\sigma \in S_C} \phi\big((T(\sigma\pi))_\pi\big) = |\mathcal{A}|$. Taking expectations conditional on $\mathcal{Z}$ and applying exchangeability to each term,

$$M \Pr(\hat{p} \le \alpha \mid \mathcal{Z}) = \mathbb{E}\big[\, |\mathcal{A}| \,\big|\, \mathcal{Z}\big] \le \lfloor \alpha M \rfloor,$$

so $\Pr(\hat{p} \le \alpha \mid \mathcal{Z}) \le \lfloor \alpha M \rfloor / M \le \alpha$. Averaging over $\mathcal{Z}$ preserves the bound.

Two cases sharpen it. When the M values are almost surely distinct, N takes each of $1, \ldots, M$ exactly once, so $|\mathcal{A}| = \lfloor \alpha M \rfloor$ and the bound holds with equality. When they fall into $M/\varkappa$ groups of exactly $\varkappa$ equal values, distinct across groups, which is what Proposition 3(ii) gives with $\varkappa = 2$ for the two-sided Wald statistic, N takes each of $\varkappa, 2\varkappa, \ldots, M$ with multiplicity $\varkappa$, so $|\mathcal{A}| = \varkappa \lfloor \alpha M / \varkappa \rfloor$ and $\Pr(\hat{p} \le \alpha) = \varkappa \lfloor \alpha M / \varkappa \rfloor / M$. In either case the randomized version rejects with probability one on $\{\hat{p} \le \alpha_-\}$, whose probability is α_-, and with probability $(\alpha - \alpha_-)/(\alpha_+ - \alpha_-)$ on $\{\hat{p} = \alpha_+\}$, whose probability is $\alpha_+ - \alpha_-$, so its size is α.

For the Monte Carlo reference set of Definition 2, replace S_C by $\mathcal{R} = \{\mathrm{id}, \pi_2, \ldots, \pi_B\}$. Because $\pi_2, \ldots, \pi_B$ are i.i.d. uniform on S_C and drawn independently of the data, the vector $(T(\pi))_{\pi \in \mathcal{R}}$ is exchangeable conditional on $\mathcal{Z}$ under Assumption 4, and the rank argument above gives $\Pr(\text{reject}) \le \alpha$ with equality at $\varkappa \lfloor \alpha B / \varkappa \rfloor / B$. □

Proof of Proposition 3. (i) Every $\pi \in \mathrm{Stab}(\delta)$ satisfies $\delta_\pi = \delta$ and hence $T(\pi) = T(\mathrm{id})$, so the numerator of $\hat{p}$ is at least $|\mathrm{Stab}(\delta)|$ on the exhaustive reference set, and $\mathrm{Stab}(\delta)$ is the direct product of the symmetric groups on the tied groups of d, of order the product of their factorials. On the Monte Carlo reference set the identity alone gives $\hat{p} \ge 1/B$.

(ii) Because the multiset $\{\delta_c\}$ is symmetric about zero, pick an involution $r \in S_C$ matching

each c with $\delta_c = v > 0$ to some c' with $\delta_{c'} = -v$ and fixing every c with $\delta_c = 0$; then $\delta_{r(c)} = -\delta_c$ for all c, and $r \neq \mathrm{id}$ because δ is not identically zero. With equally spaced d we may write $\delta_c = s\,(c - \frac{C+1}{2})$ for a scale $s > 0$ and take $r(c) = C + 1 - c$. Pre-composition negates the label vector: $(\delta_{r\circ\pi})_c = \delta_{r(\pi(c))} = -\delta_{\pi(c)}$, so $\delta_{r\circ\pi} = -\delta_\pi$. Evenness of $\mathcal{T}$ then gives $T(r\circ\pi) = T(\pi)$ for every π. The map $\pi \mapsto r\circ\pi$ is left translation by an involution other than the identity, hence a fixed-point-free involution on S_C, so the M values fall into $M/2$ equal pairs, and $r \circ \mathrm{Stab}(\delta)$ is disjoint from $\mathrm{Stab}(\delta)$ because $r \notin \mathrm{Stab}(\delta)$. The numerator of $\hat{p}$ is therefore at least $2|\mathrm{Stab}(\delta)|$.

For the Wald statistic, $\hat{\gamma}(-\delta) = -\hat{\gamma}(\delta)$ by Lemma 2, each cluster score is odd in δ, and the CR1 covariance is quadratic in the cluster scores, so $\hat{V}_\gamma(-\delta) = \hat{V}_\gamma(\delta)$ and the quadratic form is unchanged.

(iii) Rejection at level α requires some attainable $\hat{p} \leq \alpha$, hence $2|\mathrm{Stab}(\delta)|/C! \leq \alpha$ in the even case and $|\mathrm{Stab}(\delta)|/C! \leq \alpha$ otherwise, which reduce to $2/C! \leq \alpha$ and $1/C! \leq \alpha$ when d has no ties. □

Proof of Theorem 2. For the first claim, replace S_C in the proof of Theorem 1 by the finite group $\mathcal{E} = \{-1, +1\}^G$ acting by coordinatewise sign change. That proof uses only finiteness of the group and invariance of the data law under it, both of which the sign-symmetry hypothesis supplies.

For the second claim, let u_{gc} denote the cluster-g category-c errors under (4.2), taken before the two-way within transformation, and build the infeasible contribution from them,

$$\zeta_g^0 = \frac{\left(S^{-1}\sum_c \delta_c \tilde{X}_g' u_{gc}\right)_P}{\sum_c \delta_c^2}.$$

Using the raw rather than the within-transformed errors matters here and costs nothing. It matters because the two-way within projection subtracts period means, which pool across clusters, so contributions formed from the within-transformed errors are dependent across g. It costs nothing because $\tilde{X}'\Pi = \tilde{X}'$, so the aggregate is unchanged and $\sum_g \zeta_g^0 = \hat{\gamma}_P - \gamma_P$ still holds. Each ζ_g^0 is then a fixed linear functional of cluster g's errors alone, so independence across g together with symmetry of each summand about zero gives $(\epsilon_g \zeta_g^0)_g \overset{d}{=} (\zeta_g^0)_g$ for every $\epsilon \in \mathcal{E}$.

For the third claim, write z_g^r for the cluster-g score of the γ block from the fit with (4.2) imposed, so the feasible contribution is $\zeta_g = (S^{-1} z_g^r)_P / \sum_c \delta_c^2$. It replaces the errors by the restricted residuals, which subtracts a term $S^{-1} S_g[\sum_h (\cdot)_h]$. The factor $\|S_g\|/\|S\|$ is $O(G^{-1})$ and it multiplies a sum of G cluster contributions, which is $O_p(G^{1/2})$ times a single one; relative to ζ_g itself the discrepancy is therefore $O_p(G^{-1/2})$.

Asymptotic level α does not follow from an approximate-symmetry argument over $\mathcal{E}$. The results of Canay et al. (2017) hold for a randomization group whose cardinality is fixed in the sample size, and $|\mathcal{E}| = 2^G$ grows with G. We argue it instead as a statement about the

conditional law of the randomization distribution. Assume (W1) the cluster blocks $(\tilde{X}_g, \{u_{gc}\}_c)$ are independent across g with $\mathbb{E}[z_g^0] = 0$ under (4.2) and uniformly bounded moments of order strictly greater than four, where $z_g^0 = \sum_c \delta_c \tilde{X}_g' u_{gc}$; (W2) $\max_g \|S_g\|/\|S\| \to 0$ and $G^{-1}S \to \bar{S} \succ 0$; and (W3) $G^{-1}\sum_g \mathbb{E}[z_g^0 z_g^{0\prime}] \to \Sigma_z \succ 0$ on the block P. The sign-change distribution is the null-imposed Rademacher wild cluster bootstrap of Cameron et al. (2008) in score form. Conditional on the data, $G^{-1/2}\sum_g \epsilon_g z_g^r$ is a sum of G independent mean-zero terms with conditional variance $G^{-1}\sum_g z_g^r z_g^{r\prime}$, which converges in probability to Σ_z under (W1) to (W3), and (W1) with (W2) gives the conditional Lyapunov condition. The conditional bootstrap central limit theorem of Djogbenou et al. (2019) then gives the randomization distribution of the statistic the same χ^2_m limit, conditionally in probability, that the statistic itself has under (4.2). The randomization p-value is therefore asymptotically uniform. □

Proof of Corollary 3. Under the alternative $\gamma = \kappa k$, Lemma 2 gives $\hat{\gamma} = \sum_c \delta_c b_c/\Sigma_\delta^2$ with $\mathbb{E}[b_c] = (\beta_\ell)_{\ell \in L} + \delta_c \kappa k$, and with homoskedastic categories independent across c, $\mathrm{Var}(b_c) = \sigma^2 S^{-1}$. Since $\sum_c \delta_c = 0$, $\hat{\gamma}$ has mean κk and covariance $(\sigma^2/\Sigma_\delta^2)S^{-1}$, so on the post-treatment block $V = \mathrm{Var}(\hat{\gamma}_P) = (\sigma^2/\Sigma_\delta^2)[S^{-1}]_{PP}$ and, by the definition of I_k in Section 8, $\kappa^2 k_P' V^{-1} k_P = \kappa^2 \Sigma_\delta^2 I_k/\sigma^2$.

The first expression in (8.2) is the minimum detectable response of the infeasible Wald statistic that uses V in place of an estimate. That statistic is noncentral chi-square with m degrees of freedom and noncentrality $\lambda = \kappa^2 \Sigma_\delta^2 I_k/\sigma^2$; setting $\lambda = \lambda^*(\alpha, \beta, m)$ and solving for κ gives it. The feasible CR1 Wald statistic does not have that law. The identity in the proof of Proposition 2 uses no distributional assumption and holds under the alternative with $\bar{\eta}_P$ replaced by $\hat{\gamma}_P/G$, so under Assumptions 2 and 3 the feasible statistic is ω_2 times a noncentral Hotelling T^2 with dimension m and $G-1$ degrees of freedom. Equation (8.2) therefore understates the minimum detectable response of the feasible test twice, through the critical value, as Section 8 records, and through the distribution under the alternative.

For the directed statistic, t is standard normal under (4.2) and normal with unit variance and mean $\kappa\sqrt{\Sigma_\delta^2 I_k}/\sigma$ under the alternative, so power $1-\beta$ at level α requires that mean to equal $z_{1-\alpha} + z_\beta$, which gives the second expression. □

Proof of Theorem 3. Two configurations $(b_c)_c$ with the same contrast vector $(b_c - \bar{b})_c$ differ by a shift common to all categories, so the $C-1$ contrasts are a maximal invariant and an invariant test is a function of them. The maximal invariant is the full contrast vector, of dimension $(C-1)q$, and not $\hat{\gamma}_P$. Decompose the contrast space orthogonally in the category index into the direction δ and its complement. The δ-component is $\hat{\gamma} = \sum_c \delta_c b_c / \sum_c \delta_c^2$, with mean κk. The remaining $C-2$ contrasts have mean zero for every κ, so they are ancillary for κ, and because Σ does not vary with c they are uncorrelated with $\hat{\gamma}$ and, being jointly Gaussian with it, independent of it. Write A for that block.

Conditional on $A = a$, the law of $\hat{\gamma}_P$ is $N(\kappa k_P, V)$ and does not depend on a. Its density

in κ is proportional to $\exp\{\kappa k_P' V^{-1}\hat{\gamma}_P - \frac{1}{2}\kappa^2 k_P' V^{-1} k_P\}$ relative to the $\kappa = 0$ density, a one-parameter exponential family with natural parameter κ and natural sufficient statistic $U = k_P' V^{-1}\hat{\gamma}_P$, which has monotone likelihood ratio in κ. By the Karlin-Rubin theorem (Lehmann and Romano 2005) the conditional test rejecting for large U is uniformly most powerful at every conditional level, and at conditional level u its power is $\Psi(u) = \Phi(\mu + \Phi^{-1}(u))$ with $\mu = \kappa\sqrt{k_P' V^{-1} k_P} > 0$.

An invariant test may still randomize its level over the ancillary block, taking conditional level $\alpha(A)$ with $\mathbb{E}[\alpha(A)] = \alpha$, and ruling that out is the remaining step. Differentiating, $\Psi'(u) = \exp\{-\mu^2/2 - \mu\Phi^{-1}(u)\}$, which is strictly decreasing in u, so Ψ is strictly concave on $(0, 1)$ and Jensen's inequality gives $\mathbb{E}[\Psi(\alpha(A))] \leq \Psi(\alpha)$, with equality if and only if $\alpha(\cdot)$ is almost surely constant. The unconditional power of an invariant level-α test is therefore at most $\Psi(\alpha)$, attained by $\alpha(\cdot) \equiv \alpha$. Standardizing U by $\sqrt{k_P' V^{-1} k_P}$ gives $t \sim N(\kappa\sqrt{k_P' V^{-1} k_P}, 1)$, from which the critical value and the stated power function follow. □

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